

Fixed-Parameter Additive Gaps for Deterministic Communication Complexity Complete Human-Level Formalisation-Facing Proof Specification

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Mathematical and verification boundary

This manuscript proves one finite statement: for every fixed positive integer k , one construction chosen before the satisfiable/unsatisfiable case produces a nonempty family of Boolean matrices with an additive communication-complexity gap k and a naming-width bound on those same matrices. The proof is relative to two explicit outside inputs: a composed finite source theorem and a balanced-family theorem with the three product forms stated in Appendix D. The expander theorem used in a proof of the source theorem is upstream of that composed input; it is not an argument of the matrix construction.

The mathematical proof and the formal proof have different status. The definitions and proof below are a complete human-level specification. Many supporting declarations have been checked in Lean, but this document does not claim that the final theorem has been checked until the finite root and its axiom audit have both succeeded. Section 15 records the precise boundary.

Abstract

Let $D(M)$ denote deterministic communication complexity. We give a promise-independent finite reduction from ordered 3-CNF formulae to a nonempty family of pairs (M_j, Y_j) . For every fixed $k \geq 1$, one cutoff is chosen before the input formula. Beyond that cutoff, satisfiability implies $D(M_j) \leq Y_j$ for at least one member, whereas unsatisfiability implies $D(M_j) = Y_j + k$ for every member. The identical matrices satisfy $\max\{1, \lceil \log_2 |\text{Rows}(M_j)| \rceil, \lceil \log_2 |\text{Cols}(M_j)| \rceil\} \leq \lfloor n/(2k) \rfloor$. Hence each side has at most $2^{\lfloor n/(2k) \rfloor}$ elements and each matrix has at most $2^{\lfloor n/k \rfloor}$ entries.

The formalisation-facing presentation fixes zero-based carriers, total constructors, all fallback branches, exact entry equations, a six-number level parameter record, separate structural and semantic hypotheses, and a single dependent recursion for the matrix sequence. The difficult lower bound is organized around one concrete protocol frontier on the actual scale-alignment matrix. Its initialization, local transitions, global mass accounting, and terminal contradiction are stated separately. No desired communication or size conclusion occurs in construction data.

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Contents

1	Introduction	4
1.1	Problem and finite result	4
1.2	Why the proof has a scale-alignment step	5
1.3	Design choices for formalisation	5
2	Mechanical contract and definition order	6
2.1	Finite carriers and notation	6
2.2	Stored data and derived assertions	6
2.3	One total branch	7
2.4	Dependency-ordered definition ledger	7
2.5	Proof obligations at every transport	8
3	Finite communication foundations	8
3.1	Protocols and exact cost	8
3.2	Interlace and exact carriers	9
3.3	Retained families and profiles	10
3.4	Naming-width conversion	10
4	The two outside mathematical inputs	11
4.1	Ordered source graphs	11
4.2	Balanced families	12
4.3	Branch neutrality	14
5	Public records and separated assertions	14
5.1	One level of public data	14
5.2	Structural checklist	15
5.3	Semantic checklist	15
5.4	Public output record	17
6	Canonical finite parameter schedule	17
6.1	Inputs to the schedule	17
6.2	Reference, alignment, and bridge widths	18
6.3	Direct schedule theorem	19
6.4	Formula-independent cutoffs	19
7	The total one-level construction	21
7.1	Stages 1–3	21
7.2	Guarded source rows	22
7.3	Raw Stage 4	22
7.4	Protected labelled duplication	23
7.5	Forced post-composition	23
8	Protected Stage 4 on the actual matrix	24
8.1	The exact protected statement	24
8.2	Dense retained members	25
8.3	Source synchronization	25
8.4	Stopping frontier and contradiction	26

8.5	Exact cost	26
8.6	The actual reference restriction	27
9	Scale alignment on the actual retained game	27
9.1	Operational target	27
9.2	Physical coordinates and reference rows	27
9.3	The concrete protocol frontier	28
9.4	Hard first-Bob nodes	29
9.5	Final owners, donors, and global mass	29
9.6	Public scale-aligned certificate	32
10	The construction-independent bridge	32
10.1	The two inherited base estimates	33
10.2	Defect bookkeeping	33
10.3	Entry at the requested depth	36
11	One-level amplifier	37
12	The one canonical matrix-and-budget recursion	38
12.1	Full-cube seed	38
12.2	Dependent state	39
12.3	Root family	39
13	Gap and size induction	39
13.1	Simultaneous gap invariant	40
13.2	Actual carrier counts	40
13.3	One scalar naming-width majorant	41
14	Assembly of the finite theorem	42
14.1	Quantifier and identity audit	42
15	Mathematical status and formal verification boundary	43
15.1	What is proved on paper	43
15.2	What a completed Lean verification must show	43
A	The concrete stopping argument	44
A.1	Reached-node data	44
A.2	The four fixed extension corollaries	45
A.3	Protected stopping	46
A.4	Scale-alignment stopping	46
B	Arithmetic gates in consumption order	46
B.1	Positivity and Stage-2 range	47
B.2	Ownership of the five quantitative rows	47
B.3	Protected loss inequalities	47
B.4	Reference-block ceiling	48
B.5	Alignment and bridge	48
B.6	Owner transfer and exceptional terminal	48
B.7	Carrier and final-size arithmetic	49

C	Dependency audit	49
C.1	Public-interface correspondence	50
D	Exact outside and inherited contracts	51
D.1	Composed source interface	51
D.2	Balanced-family interface	51
D.3	Inherited communication lemmas	52
E	Adversarial proof audit	52

1 Introduction

1.1 Problem and finite result

Alice receives a row x of a finite Boolean matrix M and Bob receives a column y . They must output $M(x, y)$ by following a deterministic binary protocol tree. The quantity $D(M)$ is the least possible worst-case number of communicated bits [3, 4]. A constant matrix has cost zero. All row and column sets in this manuscript are finite and nonempty whenever D is used.

The construction starts from an ordered 3-CNF formula Φ . Its order is part of the input: clauses, the three literal positions in each clause, and repeated literals are not quotiented. Write $n(\Phi)$ for its declared number of variables and $\text{SAT}(\Phi)$ for ordinary satisfiability.

For natural numbers n, k , define

$$e(n, k) := \left\lfloor \frac{n}{2k} \right\rfloor, \quad S(n, k) := 2^{e(n, k)}, \quad E(n, k) := S(n, k)^2. \quad (1)$$

For a nonempty finite matrix $M : X \times Y \rightarrow \{0, 1\}$, define its *naming width*

$$w(M) := \max \{1, \lceil \log_2 |X| \rceil, \lceil \log_2 |Y| \rceil\}. \quad (2)$$

The logarithmic ceilings in (2) are the natural-number ceiling logarithms; thus $|X| \leq 2^{\lceil \log_2 |X| \rceil}$, including when $|X| = 1$.

Theorem 1.1 (Finite fixed-parameter gap and same-matrix size). *Fix a composed finite source theorem and a balanced-family theorem satisfying the exact contracts in Appendix D. There is one branch-neutral construction with the following property. For every $k \in \mathbb{N}$ with $k \geq 1$, there is an integer $U_k \geq k$, chosen before the formula, such that every valid ordered 3-CNF formula Φ with $n(\Phi) \geq U_k$ produces an integer $t(\Phi) > 0$ and a family*

$$(M_{\Phi, k, j}, Y_{\Phi, k, j}), \quad j \in [t(\Phi)].$$

Every $M_{\Phi, k, j}$ has nonempty row and column carriers. The family is fixed without inspecting whether Φ is satisfiable, and it satisfies

$$\text{SAT}(\Phi) \implies \exists j \in [t(\Phi)] \quad D(M_{\Phi, k, j}) \leq Y_{\Phi, k, j}, \quad (3)$$

$$\neg \text{SAT}(\Phi) \implies \forall j \in [t(\Phi)] \quad D(M_{\Phi, k, j}) = Y_{\Phi, k, j} + k, \quad (4)$$

$$\forall j \in [t(\Phi)] \quad w(M_{\Phi, k, j}) \leq e(n(\Phi), k). \quad (5)$$

For the same matrices appearing in (3)–(5),

$$|\text{Rows}(M_{\Phi, k, j})| \leq S(n(\Phi), k), \quad |\text{Cols}(M_{\Phi, k, j})| \leq S(n(\Phi), k), \quad (6)$$

$$|\text{Rows}(M_{\Phi, k, j})| |\text{Cols}(M_{\Phi, k, j})| \leq E(n(\Phi), k) \leq 2^{\lfloor n(\Phi)/k \rfloor}. \quad (7)$$

The quantifier order in [Theorem 1.1](#) is part of the statement. In particular, U_k is not allowed to depend on Φ or on the branch j , and the matrix in the size estimate is not a padded or replacement matrix.

1.2 Why the proof has a scale-alignment step

The one-bit construction inherited from the exact hardness proof is not directly iterable. The generic two-copy bridge needs two lower bounds at one small common row density. The raw fourth-stage matrix does not have such a full-cost bound. Indeed, an off-diagonal column restriction keeps at least $7/8$ of its columns while making every source entry neutral, and if an optimal protocol begins with Alice, one of its two children keeps at least half the rows at one bit less cost. These are structural obstructions, not a defect of constants.

The construction therefore performs two necessary repairs. First it makes labelled duplicate reservoirs, producing a protected matrix N with a dense full-cost estimate and three one-bit-down reference estimates. Second it interlaces q_{al} copies and transposes, producing a matrix W for which the estimates have one common threshold. A general bridge then adds one bit and restores the same robustness property required by the next level.

Dependency-ordered proof spine

The proof uses the following seven nodes.

- F1. finite matrices, protocols, restrictions, interlaces, and profiles;
- F2. a six-number parameter record and its finite inequalities;
- F3. one total fourth-stage construction and its forced post-composition;
- F4. the protected and scale-aligned lower bound on that actual object;
- F5. one generic one-level amplifier;
- F6. one dependent recursion proving the k -level gap;
- F7. one scalar majorant proving the naming-width and entry bounds.

The composed source theorem and the three product-balancedness statements are outside inputs. Every other arrow is proved in this manuscript.

1.3 Design choices for formalisation

Four choices prevent the statement from becoming accidentally weaker or vacuous.

- D1.** Every carrier is zero-based or is displayed as a sum, product, function, or subtype of previously defined finite carriers.
- D2.** Construction data contain no satisfiability witness, no protocol lower bound, and no target theorem. Structural failure selects a fixed all-false object on the already determined carriers.
- D3.** The protected matrix, its reference restriction, the scale-aligned matrix, the bridge output, and the iterated matrix are connected by unconditional equations, not by existential isomorphisms chosen later.

D4. The size induction reads the actual row and column counts generated by the matrix recursion. It does not quantify over an arbitrary numerical sequence that could be unrelated to those matrices.

The paper proof and the Lean proof need not have the same internal lemma boundaries. They do, however, use the same external assumptions, public objects, branch order, additive gap, and same-object size conclusion.

2 Mechanical contract and definition order

This section fixes conventions that are easy to leave implicit on paper but cannot be left implicit in a formal proof.

2.1 Finite carriers and notation

Convention 2.1 (Zero-based finite carrier). For $m \in \mathbb{N}$, write $[m] := \{0, \dots, m-1\}$. If $m = 0$ this set is empty. An expression $i \in [m]$ therefore carries the proof obligation $i < m$. Products, sums, function spaces, and subtypes use their literal finite meanings.

For a finite matrix $M : X \times Y \rightarrow \{0, 1\}$, write $\text{Rows}(M) = X$ and $\text{Cols}(M) = Y$. If $A \subseteq X$ and $B \subseteq Y$, then $M|_{A \times B}$ is the matrix on the subtype carriers $\{x : X : x \in A\}$ and $\{y : Y : y \in B\}$. This restriction is defined even when a subtype is empty, but every use of communication complexity separately proves nonemptiness.

All logarithms without a displayed base are natural-number base-two logarithms at positive powers of two. We distinguish $\log_2 m = \max\{r : 2^r \leq m\}$ from $\text{clog}_2 m = \min\{r : m \leq 2^r\}$. They are identified only after proving that m is a positive power of two.

2.2 Stored data and derived assertions

The following four kinds of object are never merged.

Kind	Permitted contents
Public data	Formula, source branch, seed matrix, natural parameters, balanced-family entries, and canonical carrier maps.
Structural assertion	Positivity, range, injectivity, width identities, and the finite inequalities needed for the displayed definitions.
Promise assertion	A satisfying colouring or a quantitative statement that every colouring violates many edges; exact seed cost and inherited seed profile on the negative branch.
Conclusion	A protocol upper bound, protocol refutation, exact communication cost, regenerated profile, or matrix-size bound.

A constructor may read only public data. A theorem may read structural and promise assertions. No conclusion is a field of a constructor, parameter record, source record, frontier state, or balanced-family record.

2.3 One total branch

Definition 2.2 (Structural legality). For a balanced-family provider $\mathcal{F}$, public level input P , and six-number parameter record π , define $\text{Legal}(\mathcal{F}, P, \pi)$ to be $\text{LevelOK}(\mathcal{F}, P, \pi)$ from [Definition 5.2](#). The guarded-source coordinate bound and the balanced-family entry bound are derived from this checklist and the provider contract; they are not additional stored conjuncts. Structural legality contains no statement about D or about a retained submatrix.

Every level object uses the same split:

$$\text{object}(\mathcal{F}, P, \pi) = \begin{cases} \text{genuineObject}(\mathcal{F}, P, \pi), & \text{Legal}(\mathcal{F}, P, \pi), \\ \text{zeroObject}(P, \pi), & \text{otherwise.} \end{cases}$$

The all-false fallback has the carriers already determined by P and π . It does not choose an inhabitant of a possibly empty type. Later arithmetic proves that every call used in [Theorem 1.1](#) takes the genuine branch. Since the test in [\(2.3\)](#) is a proposition and both successful proofs select the same displayed object, proof irrelevance makes the result independent of the proof term.

2.4 Dependency-ordered definition ledger

Each row below is defined before its first mathematical use. The section column is also the intended formalisation dependency order.

ID	Object	Direct dependencies
F0.1	$[m]$, finite matrix, transpose, reindex, restriction	natural numbers and finite sets
F0.2	D , bracket family, dense minimum, three-rung profile	F0.1 and the finite protocol model
F0.3	$w(M), e(n, k), S(n, k), E(n, k)$	F0.1 and ceiling logarithms
F1.1	ordered formula, source branch, common/positive/negative predicates	F0.1
F1.2	balanced family and its three product forms	F0.1
F2.1	public level input P , level parameters π , semantic scalars g	F0 and F1
F2.2	structural and semantic checklists	F2.1
F3.1	Stages 1–3 trace T	F2 and balanced-family entries
F3.2	raw matrix B and protected matrix A	F3.1
F3.3	$N, N_{\text{ref}}, U, W, R^+$ and canonical successor	F3.2
F4.1	concrete retained-game frontier state and invariant	F2–F3

ID	Object	Direct dependencies
F4.2	protected and scale-aligned certificate	F4.1
F5.1	one-level theorem and overhead H	F3.3, F4.2, bridge theorem
F6.1	full-cube seed, dependent state, matrix recursion, budget recursion	F1 and F5.1
F6.2	actual carrier majorant	F3.3 and F6.1
F7.1	root family and public output pairs	F1, F6.1, F6.2

2.5 Proof obligations at every transport

Whenever a matrix is transposed, restricted, duplicated, interlaced, or reindexed, the proof records five items: the source and target carrier types, the entry equation, the two carrier cardinalities, nonemptiness, and the communication/profile transport used later. Equality of cardinalities is not treated as equality of carrier types. Canonical zero-based presentation is always by a displayed equivalence.

The proof also keeps natural and real inequalities separate. Every division used in a real-valued density estimate is accompanied by positivity of its denominator. Every natural subtraction is justified by the preceding lower bound; it is never silently read as integer subtraction.

3 Finite communication foundations

3.1 Protocols and exact cost

A deterministic protocol over $X \times Y$ is a finite rooted binary tree. Every internal node is labelled either by a Boolean function of Alice's input or by a Boolean function of Bob's input; its two outgoing edges are labelled 0 and 1. Leaves are labelled by output bits. The protocol computes $M : X \times Y \rightarrow \{0, 1\}$ when every input pair reaches a leaf labelled $M(x, y)$. Its cost is the maximum root-to-leaf number of internal nodes. Then

$$D(M) := \min \{ \text{cost}(P) : P \text{ computes } M \}.$$

The carriers used with (3.1) are finite and nonempty, so the set of achievable natural-number costs is nonempty: Alice may identify her row and Bob may identify his column. Its minimum exists by well-ordering.

A protocol is *reduced* if every reached internal node carries a nonempty rectangle on which its remaining function is nonconstant. Deleting unreachable branches, contracting unary nodes, and replacing a monochromatic subtree by its output leaf reduces any protocol without increasing its depth.

Lemma 3.1 (Fixed-output leaf and rank bounds). *Let P be a nonconstant reduced protocol of depth at most $d \geq 1$. For each $\beta \in \{0, 1\}$, at most 2^{d-1} leaves of P have output β . Consequently, if a nonconstant Boolean matrix M satisfies*

$$\text{rank}_{\mathbb{R}}(\mathbf{1}_{M=\beta}) > 2^{d-1}, \tag{8}$$

then $D(M) \geq d + 1$.

Proof. For the leaf count, induct on d . At depth one, the two root children cannot have the same output. At greater depth, apply the induction hypothesis to each nonconstant child. A constant child contributes at most one β -leaf, so the only additional inequality is $1 + 2^{d-2} \leq 2^{d-1}$. If both children are constant, their outputs differ by reducedness.

For the rank claim, reduce an alleged depth- d protocol. Its β -leaves partition the β -entries into rectangles. Each nonempty rectangle indicator has real rank one, so rank subadditivity bounds the left side of (8) by the number of β -leaves. This contradicts the first part. $\square$

Lemma 3.2 (Elementary transports). *Let $M : X \times Y \rightarrow \{0, 1\}$ have finite carriers.*

- (a) $D(M^\top) = D(M)$.
- (b) Relabelling either carrier by a bijection preserves $D(M)$.
- (c) If $A \subseteq X$ and $B \subseteq Y$ are nonempty, then $D(M|_{A \times B}) \leq D(M)$.
- (d) Duplicating equal rows or equal columns preserves $D(M)$ provided at least one representative of every original row or column remains.

Proof. For (a), exchange the two parties at every internal node. For (b), precompose each query with the relevant bijection and use its inverse for the reverse inequality. For (c), restrict every query to the selected subtype. For (d), compose a protocol with the forgetful map for the upper bound and restrict to one representative of every original input for the lower bound. $\square$

3.2 Interlace and exact carriers

For $p \in \mathbb{N}$ and $M : X \times Y \rightarrow \{0, 1\}$ define

$$\text{Int}_p(M) : ([p] \times X) \times ([p] \rightarrow Y) \rightarrow \{0, 1\}, \quad \text{Int}_p(M)((i, x), f) = M(x, f(i)). \quad (9)$$

Thus

$$|\text{Rows}(\text{Int}_p(M))| = p|X|, \quad |\text{Cols}(\text{Int}_p(M))| = |Y|^p. \quad (10)$$

If p is a positive power of two, a selector protocol first communicates the $\log_2 p$ bits of i and then runs a protocol for M . Hence

$$D(\text{Int}_p(M)) \leq D(M) + \log_2 p. \quad (11)$$

The analogous bound with $\text{clog}_2 p$ holds for arbitrary positive p .

The two actual outer objects used later are

$$U := \text{Int}_{q_{\text{al}}}(N), \quad W := U^\top, \quad (12)$$

$$R^+ := \text{Int}_{Q_{\text{br}}}(W)^\top. \quad (13)$$

Consequently, if $m_N = |\text{Rows}(N)|$ and $n_N = |\text{Cols}(N)|$, then

$$|\text{Rows}(W)| = n_N^{q_{\text{al}}}, \quad |\text{Cols}(W)| = q_{\text{al}}m_N, \quad (14)$$

$$|\text{Rows}(R^+)| = (q_{\text{al}}m_N)^{Q_{\text{br}}}, \quad |\text{Cols}(R^+)| = Q_{\text{br}}n_N^{q_{\text{al}}}. \quad (15)$$

3.3 Retained families and profiles

For finite X, Y , an integer $p \geq 1$, and real $0 < x, z \leq 1$, let $\mathcal{B}(X, Y; p, x, z)$ be the set of pairs (A, B) with

$$\begin{aligned} A &\subseteq [p] \times X, & |\{x \in X : (i, x) \in A\}| &\geq \lceil x|X| \rceil \quad \text{for every } i \in [p], \\ B &\subseteq ([p] \rightarrow Y), & |B| &\geq \lceil z|Y|^p \rceil. \end{aligned}$$

For $M : X \times Y \rightarrow \{0, 1\}$, define

$$\mathcal{D}_M(p; x, z) := \min_{(A, B) \in \mathcal{B}(X, Y; p, x, z)} D(\text{Int}_p(M)|_{A \times B}). \quad (16)$$

The row threshold is per outer block, not merely a bound on the sum over all blocks. The row carrier is nonempty whenever $p \geq 1$, $x > 0$, and X is nonempty. The column carrier is nonempty whenever $z > 0$ and Y is nonempty. Taking all rows and columns proves that the bracket family itself is nonempty when $x, z \leq 1$; these facts are proved before taking the minimum.

Define the three-rung profile

$$\Lambda_M(p; x, z) := \min \{ \mathcal{D}_M(p; x, z), 1 + \mathcal{D}_M(p; x, z/2), 2 + \mathcal{D}_M(p; x, z/4) \}. \quad (17)$$

For a natural depth b , put $x_b = 2^{-b}$ and

$$\text{Rob}(M; b, z) := \Lambda_M(1; x_b, z). \quad (18)$$

The inequality $C \leq \text{Rob}(M; b, 3/5)$ expands to the following three separate statements:

$$\begin{aligned} C &\leq \mathcal{D}_M(1; x_b, 3/5), \\ C - 1 &\leq \mathcal{D}_M(1; x_b, 3/10), \\ C - 2 &\leq \mathcal{D}_M(1; x_b, 3/20). \end{aligned} \quad (19)$$

The subtractions in (19) are used only when the relevant lower bounds on C have already been proved.

Lemma 3.3 (Reindexing preserves retained minima). *Relabelling both carriers by bijections preserves $\mathcal{D}_M(p; x, z)$, $\Lambda_M(p; x, z)$, and $\text{Rob}(M; b, z)$.*

Proof. The bijections induce cardinality-preserving bijections between the two bracket families. Each corresponding restricted matrix is a relabelling of the other, so Lemma 3.2(b) applies pointwise. Taking minima and then the displayed three-term minimum preserves equality. $\square$

3.4 Naming-width conversion

Lemma 3.4 (Same-object size conversion). *Let $k \geq 1$ and let M be a nonempty finite matrix. If $w(M) \leq e(n, k)$, then*

$$\begin{aligned} |\text{Rows}(M)| &\leq S(n, k), & |\text{Cols}(M)| &\leq S(n, k), \\ |\text{Rows}(M)| |\text{Cols}(M)| &\leq E(n, k) \leq 2^{\lfloor n/k \rfloor}. \end{aligned}$$

Proof. The definition of w gives both ceiling logarithms at most $e(n, k)$, hence both side bounds. Multiplication gives the first entry bound. Finally, natural-number division satisfies

$$2 \left\lfloor \frac{n}{2k} \right\rfloor \leq \left\lfloor \frac{n}{k} \right\rfloor,$$

because $e(n, k)(2k) \leq n$ and $k > 0$. Monotonicity of 2^r gives the last inequality. Every line refers to the original carriers of M . $\square$

4 The two outside mathematical inputs

This section states the complete outside boundary. Neither input contains a matrix produced by the reduction or a conclusion about communication complexity.

4.1 Ordered source graphs

An *ordered multigraph* G consists of integers $v(G), e(G)$ and two endpoint maps

$$\partial_0, \partial_1 : [e(G)] \longrightarrow [v(G)].$$

An edge occurrence is its index in $[e(G)]$; parallel occurrences remain distinct. A loop contributes two incidence positions to degree. For a three-colouring $c : [v(G)] \rightarrow [3]$, let

$$\text{mono}(G, c) := |\{r \in [e(G)] : c(\partial_0 r) = c(\partial_1 r)\}|.$$

A source branch is a triple $S = (G, \gamma, \Delta)$ with rational $0 < \gamma \leq 1$ and positive integer Δ . Define:

$$\text{Common}(S) : \Longleftrightarrow \begin{cases} v(G), e(G) > 0, \\ \partial_0 r \neq \partial_1 r & \text{for every occurrence } r, \\ 1 \leq \deg_G(u) \leq \Delta & \text{for every vertex } u; \end{cases} \quad (20)$$

$$\text{SourceYes}(S) : \Longleftrightarrow \exists c : [v(G)] \rightarrow [3] \quad \text{mono}(G, c) = 0; \quad (21)$$

$$\text{SourceNo}(S) : \Longleftrightarrow \forall c : [v(G)] \rightarrow [3] \quad \gamma e(G) \leq \text{mono}(G, c). \quad (22)$$

Assumption 4.1 (Composed finite source theorem). There are fixed integers $c_{\text{src}}, \Delta, n_{\text{src}} \geq 1$, a fixed rational $0 < \gamma \leq 1$, and for every ordered 3-CNF formula Φ a finite ordered list $\mathcal{S}(\Phi)$ of graphs. For every valid Φ with $n(\Phi) \geq n_{\text{src}}$:

(S1) $t(\Phi) := |\mathcal{S}(\Phi)|$ is positive;

(S2) each indexed branch $S_{\Phi, j} = (\mathcal{S}(\Phi)_j, \gamma, \Delta)$ satisfies $\text{Common}(S_{\Phi, j})$;

(S3) $n(\Phi) \leq e(\mathcal{S}(\Phi)_j)$ and

$$v(\mathcal{S}(\Phi)_j) + e(\mathcal{S}(\Phi)_j) \leq n(\Phi)(\log_2(n(\Phi) + 2))^{c_{\text{src}}};$$

(S4) if Φ is satisfiable, some indexed branch satisfies SourceYes ;

(S5) if Φ is unsatisfiable, every indexed branch satisfies SourceNo .

The same ordered list is used in (S4) and (S5).

The theorem in [Assumption 4.1](#) is already composed; its intended source is the companion construction [2]. A proof of it may use a finite expander theorem to normalize the source graphs, but the downstream construction receives only the list $\mathcal{S}(\Phi)$ and the five displayed properties. In particular, there is no independent expander parameter in [Theorem 1.1](#).

Lemma 4.2 (Integer-logarithm normalization of the source). *The source input can be replaced by one with exactly the same ordered graph lists, γ , and Δ , satisfying*

$$v(G) + e(G) \leq n \lceil \log_2(n + 2) \rceil^{\hat{c}_{\text{src}}}, \quad \hat{c}_{\text{src}} = 2c_{\text{src}}, \quad \hat{n}_{\text{src}} = \max\{n_{\text{src}}, 2\}.$$

All five source clauses remain valid beyond the new cutoff.

Proof. For $n \geq 2$, put $L = \lfloor \log_2(n+2) \rfloor$. Then $L \geq 2$, and

$$0 \leq \log_2(n+2) < L+1 \leq L^2.$$

Raising to the nonnegative integer power c_{src} and multiplying by n proves the new bound. The other clauses concern unchanged graphs; raising the cutoff only restricts their domain of application. $\square$

We use this normalized provider before selecting any construction parameters. Below, $c_{\text{src}}, n_{\text{src}}$ denote its constants again. Thus the finite implementation's integer logarithm does not require a stronger outside source theorem.

4.2 Balanced families

For positive m and integers $1 \leq t \leq q$, a finite indexed family is

$$\mathcal{F} = (f_\ell : [q] \rightarrow [m])_{\ell \in [N]}.$$

For $J \subseteq [q]$ and $a : J \rightarrow [m]$, let

$$C_{\mathcal{F}}(J, a) := |\{\ell \in [N] : f_\ell|_J = a\}|.$$

The indices in J are distinct because J is a set. Family indices are counted separately even when two functions f_ℓ coincide.

We shall also need the exact predicate used for the three product clauses. If $\mathcal{C} = (c_\ell : [Q] \rightarrow \mathcal{Y})_{\ell \in [L]}$ is an indexed family over a nonempty finite alphabet $\mathcal{Y}$, write $\text{Bal}_t(\mathcal{C}; \epsilon)$ when $L > 0$ and, for every $J \subseteq [Q]$ with $r = |J| \leq t$ and every total pattern $a : [Q] \rightarrow \mathcal{Y}$,

$$\left| \frac{|\{\ell \in [L] : c_\ell(\alpha) = a(\alpha) \text{ for every } \alpha \in J\}|}{L} - \frac{1}{|\mathcal{Y}|^r} \right| \leq \frac{\epsilon}{|\mathcal{Y}|^r}. \quad (23)$$

The pattern is total only to avoid a dependent function type; the count reads it on J only.

For a finite donor set D , define the literal donor product $\mathcal{C}^{\times D}$ as follows. Its family indices are the functions $\lambda : D \rightarrow [L]$, its coordinates are pairs $(d, \alpha) \in D \times [Q]$, and

$$\mathcal{C}_\lambda^{\times D}(d, \alpha) = c_{\lambda(d)}(\alpha). \quad (24)$$

Fixed cardinality-preserving bijections identify these two carriers with $[L^{|D|}]$ and $[|D||Q|]$. Thus (24) is a definition, not an unspecified product operation.

Assumption 4.3 (Finite balanced-family theorem). For every q, t, m with $m > 0$ and $1 \leq t \leq q$, there is one indexed family $\mathcal{F}(q, t, m)$ with $N > 0$, all entries in $[m]$, and

$$N < (2 \max\{q, 1000mt^2, 2\})^t. \quad (25)$$

For every $J \subseteq [q]$ with $r = |J| \leq t$ and every pattern $a : J \rightarrow [m]$,

$$(100t - 1)N \leq 100t m^r C_{\mathcal{F}}(J, a) \leq (100t + 1)N. \quad (26)$$

In addition, the following three product forms hold for the actual family selected by the parameter schedule. At one canonical level let

$$\mathcal{C} = (c_{\ell, \alpha})_{\ell \in [N_2], \alpha \in [q_2]}$$

be the actual Stage-2 family row with alphabet X_1 , and put

$$\epsilon_2 = \frac{1}{100t_2}, \quad \epsilon_{\text{prod}} = \max \left\{ (1 + \epsilon_2)^{t_2} - 1, 1 - (1 - \epsilon_2)^{t_2} \right\}. \quad (27)$$

For $e \geq 1$, $J' \subseteq [2^e]$, and $|J'| = 2^e/2 = 2^{e-1}$, use the subtype $D_{e,J'} = J'$ as the donor set. Then the three fields are exactly:

(B1) for every integer $e \geq 1$ and every such J' ,

$$\text{Bal}_{t_2}(\mathcal{C}^{\times D_{e,J'}}; \epsilon_2);$$

(B2)

$$\text{Bal}_{t_2}(\mathcal{C}^{\times [2]}; \epsilon_{\text{prod}});$$

(B3) for the canonical exceptional-terminal exponent $e_{\text{exc}} \geq 2$, and for every $J' \subseteq [2^{e_{\text{exc}}}]$ of cardinality $2^{e_{\text{exc}}}/2$,

$$\text{Bal}_{t_2}(\mathcal{C}^{\times D_{e_{\text{exc}},J'}}; \epsilon_2).$$

At a use site the coordinate carrier $D_{e,J'} \times [q_2]$ is reindexed by the displayed fixed equivalence in Appendix D; reindexing changes none of the counts in (23). Clauses (B1)–(B3) assert only counts in finite product families. They contain no protocol, cost, or terminal conclusion.

The entire input, including the product clauses, has the following explicit realization. Its proof remains outside Lean, as does the balanced-family existence theorem itself. Small almost-independent families [1] provide background; we use the elementary construction and exact estimates below, not an unstated strengthening of (26).

Theorem 4.4 (One family realizing every product clause). *For every legal (q, t, m) , there is a family satisfying (25) such that, for every finite donor set D ,*

$$\text{Bal}_t(\mathcal{F}(q, t, m)^{\times D}; 1/(100t)).$$

The same chosen family satisfies the base contract and all of (B1)–(B3).

Proof. Put $X = \max\{q, 1000mt^2, 2\}$. Bertrand's postulate supplies a prime p with $X < p < 2X$; fix the least such prime. Identify $[q]$ with the distinct residues $0, \dots, q-1$ in $\mathbb{F}_p$. Index the family by the p^t coefficient vectors of polynomials of degree less than t , including leading zero coefficients. Evaluate a polynomial at the selected coordinate, take the representative in $[p]$, and reduce it modulo m . This fixes all entries without a source promise and gives $N = p^t < (2X)^t$.

For $a \in [m]$, let b_a be the number of representatives in $[p]$ congruent to a modulo m . The buckets partition $[p]$, and $b_a \in \lfloor p/m \rfloor, \lceil p/m \rceil$. With $\delta = m/p$,

$$1 - \delta \leq \frac{mb_a}{p} \leq 1 + \delta, \quad 0 < \delta < \frac{1}{1000t^2}. \quad (28)$$

Evaluation at any $s \leq t$ distinct coordinates is a surjective linear map from $\mathbb{F}_p^t$ to $\mathbb{F}_p^s$: polynomial interpolation gives surjectivity, and each fibre has p^{t-s} elements. Consequently a uniform family index has independent uniform field values on those coordinates.

Now prescribe values on a set $J \subseteq D \times [q]$ of cardinality $r \leq t$. For each donor d , there are $r_d \leq t$ distinct prescribed coordinates. Applying the preceding evaluation count within that donor, and multiplying across independently chosen donor indices, gives the exact identity

$$\frac{m^r C_{\mathcal{F} \times D}(J, a)}{N^{|D|}} = \prod_{(d,j) \in J} \frac{mb_{a(d,j)}}{p}. \quad (29)$$

Donors with no prescribed coordinate contribute 1. Thus it is the total number r of prescribed coordinates, not $|D|$, that controls the error. The identity also covers $J = \emptyset$ and $D = \emptyset$.

For completeness, set $x = r\delta$. Then $0 \leq x \leq 1/(1000t) < 1$. Bernoulli's inequality and the binomial expansion give

$$(1 - \delta)^r \geq 1 - r\delta, \quad (1 + \delta)^r \leq \sum_{i=0}^r (r\delta)^i \leq \frac{1}{1 - r\delta}.$$

In the upper bound, $\binom{r}{i} \leq r^i$; the case $r = 0$ is immediate. In particular,

$$r\delta \leq \frac{1}{1000t} \leq \frac{1}{100t}, \quad \frac{r\delta}{1 - r\delta} \leq \frac{1}{1000t - 1} \leq \frac{1}{100t}.$$

Combining these inequalities with (28)–(29) proves the asserted balancedness. The product index carrier is nonempty since $N > 0$, even when D is empty. Bijective relabelling of the alphabet, family indices, or coordinate carrier preserves the counts.

A singleton donor gives the base contract. The arbitrary donor set gives (B1) and (B3) directly, uniformly in their exponents and subsets. For (B2), Bernoulli's inequality gives $(1 + \epsilon_2)^{t_2} - 1 \geq t_2\epsilon_2 \geq \epsilon_2$; hence $\epsilon_{\text{prod}} \geq \epsilon_2$, and the stronger bound just proved implies the required one. The fixed prime and coefficient ordering select one provider for all triples, before any protocol or promise. On invalid parameter triples use the singleton all-zero family, so the provider is total; its specification only asserts facts for legal requests. $\square$

This is an existence and counting proof for the selected polynomial family. It does not assert that the coarse base estimate alone forces (B1) or (B3) for every other family satisfying it.

4.3 Branch neutrality

The source list and every balanced family in this section are fixed before a source promise is invoked. The positive source implication chooses a branch from the fixed list. The negative implication proves a statement for all branches in that same list. No construction below calls a different source or family according to the promise.

5 Public records and separated assertions

5.1 One level of public data

Fix a source branch $S = (G, \gamma, \Delta)$. A public level input is

$$P = (S, m_R, n_R, R), \quad R : [m_R] \times [n_R] \rightarrow \{0, 1\}. \quad (30)$$

The record stores neither $D(R)$ nor a promise bit. Positivity of m_R, n_R is a proposition proved at each use.

Definition 5.1 (Six-number level record). The level constructor reads exactly

$$\pi = (a, s, q_2, t_2, q_{\text{al}}, Q_{\text{br}}) \in \mathbb{N}^6.$$

The first four numbers define Stages 1–4; q_{al} defines the scale-alignment interlace; Q_{br} defines the generic bridge interlace. The symbols $q_2, q_{\text{al}}, Q_{\text{br}}$ are distinct. Only a proved equation may identify two width expressions.

The proof also names

$$g = (D_0, b_0, b_+, k_{\text{ref}}, b_T, u_T, \eta_R, \eta_C, \delta_U). \quad (31)$$

These are fixed numbers, not assertions. They record the input cost and depth names, the output depth, the reference exponent and depth, a terminal width, and three deletion parameters. The exact cost targets are derived:

$$C(P, \pi, g) := D_0 + a + \log_2 q_2 + \log_2 s + 1, \quad (32)$$

$$C_W(P, \pi, g) := C(P, \pi, g) + \log_2 q_{\text{al}}, \quad (33)$$

$$H(P, \pi, g) := a + \log_2 q_2 + \log_2 s + \log_2 q_{\text{al}} + \text{clog}_2 Q_{\text{br}}. \quad (34)$$

5.2 Structural checklist

Definition 5.2 (Level checklist). For the balanced-family provider $\mathcal{F}$, define $\text{LevelOK}(\mathcal{F}, P, \pi)$ to be the conjunction

(L1) $v(G) > 0$, $m_R > 0$, and $n_R > 0$;

(L2) $a \geq 2$ and $s \geq 4$;

(L3) $q_2 = 2^{\text{clog}_2(12s e(G))}$;

(L4) $1 \leq t_2$ and $4t_2 \leq q_2$;

(L5) in each guarded-source column, every active source row has its zero-based active rank below s ;

(L6) $q_{\text{al}} > 0$ and $Q_{\text{br}} > 0$.

Item (L5) is a statement about the displayed finite map, not a stored list of ranks. Two facts deliberately are not extra fields. First, the definition of q_2 proves that every guarded-source coordinate rank is below q_2 . Second, the range clause of the base balanced-family contract proves that every selected entry lies in the Stage-1 row carrier. The checklist and those two derived facts imply positivity of every core and post-core carrier used by the genuine branch.

5.3 Semantic checklist

The negative proof separates the current seed facts from scalar facts.

Definition 5.3 (Negative seed facts). $\text{NoSeed}(P, g)$ means

(N1) $\text{Common}(S)$ and $\text{SourceNo}(S)$;

(N2) $D_0 \geq 3$;

(N3) $D(R) = D_0$;

(N4) $D_0 \leq \text{Rob}(R; b_0, 3/5)$.

Let $\zeta = \gamma/24$, let

$$\delta_V = \frac{\gamma e(G)}{2\Delta v(G)}, \quad \mu = \frac{1}{184}, \quad \tau = 2^{-21}, \quad \nu = 2^{-22}.$$

Define m_{HS} as the least integer $r \geq 5$ such that, for every integer $m \geq r$, every finite nonempty Boolean matrix M , and every real $b \geq 1$ with $m \leq b$ and $3 \leq D(M) \leq \text{Rob}(M; b, 3/5)$, one has

$$\mathcal{D}_M \left(\frac{9}{16} 2^m; 2^{m-b}, 2^{-2^{(49/100)\sqrt{m}}} \right) \geq D(M) + m. \quad (35)$$

The hard-seed lemma of [2], specialized to copy-depth parameter 5 and margin 1/10, proves that such a uniform threshold exists. Fix this least threshold once and for all, before the matrix or source input is chosen. It is an exponent; its associated width is $2^{m_{\text{HS}}}$. For the actual family selected at this level, also put

$$\begin{aligned} q_1 &= 2^a - 2, & N_2 &= |\mathcal{F}(q_2, t_2, q_1 m_R)|, \\ T_0 &= s N_2, & V_0 &= 4s v(G) m_R, & B_0 &= 4s(s-3) m_R. \end{aligned} \quad (36)$$

These are the literal template, vertex-source, and blocker-source row counts used in Stage 4; Section 7 defines their carriers and proves the same cardinality equations.

The full scalar checklist $\text{ScalarOK}(\mathcal{F}, P, \pi, g)$ consists of the following groups. Appendix B prints their canonical producers.

Raw upper protocol.

$$s = 2^{\log_2 s}, \quad s \leq 2^a. \quad (37)$$

Protected loss. The numbers η_R, η_C lie in $(0, 1]$, $u_T \leq q_{\text{al}}$, and

$$8(2^a - 2) \leq t_2, \quad (38)$$

$$\frac{64s(2^a - 2)}{\zeta} \leq q_2, \quad (39)$$

$$\eta_R < \frac{(1/4)\mu\zeta}{8s}, \quad (40)$$

$$\eta_C \leq \frac{1}{2} \min \left\{ \frac{\tau}{1000s}, \frac{\nu\delta_V}{16}, \frac{\nu}{16(s-3)} \right\}. \quad (41)$$

Template dominance. The actual Stage-4 row reservoirs satisfy

$$100(V_0 + B_0) \leq T_0. \quad (42)$$

This is a scalar count, not a statement about a retained matrix.

Reference rungs. The exact elementary reference gates are

$$\begin{aligned} 1 &\leq k_{\text{ref}}, & m_{\text{HS}} + 2 &\leq \log_2 s, & \log_2 s &< b_T, \\ k_{\text{ref}} + 2 &\leq \log_2 q_{\text{al}}, & q_{\text{al}} 2^{-b_T} &\leq \frac{4}{25025}, & a + 2 + \log_2 s &\leq b_0. \end{aligned} \quad (43)$$

For $r = 0, 1, 2$, the explicit retained-template calculation yields at least $9s/(16 \cdot 2^r)$ reference blocks after deleting the prescribed $s2^{-r-b_T}$ row fraction. This is the ceiling inequality printed in Appendix B.

Scale alignment and bridge.

$$q_{\text{al}} = 2^{\log_2 q_{\text{al}}}, \quad \delta_U = \frac{\eta_R}{16q_{\text{al}}} \leq \frac{1}{64}, \quad (44)$$

$$Q_{\text{br}} = \text{Qwidth}(b_+, \delta_U), \quad \theta := \frac{1}{160Q_{\text{br}}}. \quad (45)$$

Here $\text{Qwidth}(b, \delta)$ is the least positive power of two satisfying $80b \leq Q\delta$. Its minimality is used only for size; its defining inequality is used by the bridge.

Definition 5.4 (Full negative checklist). $\text{NoLevelOK}(\mathcal{F}, P, \pi, g)$ is $\text{NoSeed}(P, g)$ together with $\text{ScalarOK}(\mathcal{F}, P, \pi, g)$. It contains no claim about A, N, W, R^+ , no retained-family lower bound, and no protocol refutation.

For the positive branch only the common source facts, (37), the scale/bridge identities, and an upper bound on $D(R)$ are used. Thus negative information is not required to construct or analyze the positive protocol.

5.4 Public output record

One final branch output is

$$O = (m, n, h_m, h_n, M, Y), \quad (46)$$

where $h_m : m > 0$, $h_n : n > 0$, $M : [m] \times [n] \rightarrow \{0, 1\}$, and $Y \in \mathbb{N}$. It stores no gap or size assertion. Its derived entry count is mn and its derived naming width is $\max\{1, \log_2 m, \log_2 n\}$.

The root constructor returns one such record for every source-list index. A fixed 1×1 all-false fallback is used only if the literal final carriers are not positive. The cutoff proof shows that this fallback is never selected in [Theorem 1.1](#).

6 Canonical finite parameter schedule

This section gives one schedule that is deliberately generous. Its purpose is to make every later inequality a direct consequence of a definition. No number below is selected from a satisfiability or unsatisfiability witness.

6.1 Inputs to the schedule

Besides P , the schedule reads

$$I = (D, Y, b_0, b_+) \in \mathbb{N}^4, \quad h \in \mathbb{N},$$

where h is a seed exponent. Define

$$\begin{aligned} \sigma &= \max\{6, h + 2\}, & s &= 2^\sigma, \\ r_2 &= \log_2(12s e(G)), & q_2 &= 2^{r_2}, \\ \ell_1 &= \log_2(r_2 + 18), & u_1 &= \ell_1 + 2, \\ a &= h + 2 + 16u_1^2, & q_1 &= 2^a - 2, \\ b_1 &= 2^{u_1}, & b_2 &= \log_2(32sq_1m_R + 2) + 6. \end{aligned} \quad (47)$$

Put $\mu = 1/184$, $\zeta = \gamma/24$, and

$$\delta_V = \frac{\gamma e(G)}{2\Delta v(G)}.$$

Then set

$$\eta_C = \frac{1}{4} \min \left\{ \frac{2^{-21}}{1000s}, \frac{2^{-22}\delta_V}{16}, \frac{2^{-22}}{16(s-3)} \right\}, \quad (48)$$

$$\eta_R = \frac{1}{4} \min \left\{ \frac{\mu}{100}, \frac{\mu\zeta}{32s}, \eta_C \right\}, \quad n_\eta = \max\{2, \lceil 1/\eta_R \rceil\}. \quad (49)$$

All denominators are positive after $h \geq 5$ and $\text{Common}(S)$ have been established.

6.2 Reference, alignment, and bridge widths

Define

$$\begin{aligned}
U_{\text{ref}} &= \max \{16, D + Y + b_0 + b_+ + r_2 + \sigma + a + n_\eta + 64\}, \\
L_{\text{ref}} &= \text{clog}_2 U_{\text{ref}}, \\
k_{\text{ref}} &= 2^{20} + 4096(L_{\text{ref}} + 1)^2, \\
r_{\text{al}} &= 64(k_{\text{ref}} + L_{\text{ref}} + \sigma + 1), \\
q_{\text{al}} &= 2^{r_{\text{al}}}, \\
\delta_U &= \frac{\eta_R}{16q_{\text{al}}}, \\
\theta &= \frac{1}{160Q_{\text{br}}}.
\end{aligned}
\tag{50}$$

The least-width definition gives

$$80b_+ \leq Q_{\text{br}}\delta_U, \quad 0 < \theta, \quad 4Q_{\text{br}}\theta = \frac{1}{40} < \frac{1}{10}. \tag{51}$$

The terminal width and t_2 are selected last because they dominate all previous finite requests. For $0 \leq \rho < 1$ and $\theta > 0$, define

$$\begin{aligned}
B(\rho, \theta) &= \min\{B \in \mathbb{N} : B \geq 1, \rho^B < \theta\}, \\
(B_{\text{ref}}, B_{\text{sing}}, B_{\text{pair}}) &= (B(18/25, \theta), B(1 - \eta_C, \theta), B(99/100, \theta)).
\end{aligned}
\tag{52}$$

The set in this minimum is nonempty because $\rho^B \rightarrow 0$. A tuple set in a finite product of nonempty sets, of density at least θ , has a coordinate set of size at most $j(B(\rho, \theta) - 1)$ coordinates meeting every j -coordinate projection whose density is below ρ (meeting its coordinate set). To prove this, take a maximal disjoint collection of deficient projections. If it contained $B(\rho, \theta)$ members, product counting would force the original density below $\rho^{B(\rho, \theta)} < \theta$. The union of the collection is the required set. The three exponents apply respectively to the small-reference, single-coordinate, and paired-coordinate projections. Put

$$\begin{aligned}
t_{\text{ref}} &= 2^{k_{\text{ref}}}, \\
e_{\text{max}} &= (B_{\text{ref}} - 1)(t_{\text{ref}}/2 + \log_2 t_{\text{ref}} - 2) + (B_{\text{sing}} - 1) + 2(B_{\text{pair}} - 1), \\
L_q &= \text{clog}_2(2q_{\text{al}}) + 1, \\
E_B &= \left\lceil \frac{512\mu B_{\text{pair}} L_q s e_{\text{max}}}{(1 - 9/16)\eta_R} \right\rceil, \\
E_T &= \left\lceil \frac{e_{\text{max}}}{1 - 9/16} \right\rceil + E_B, \\
\delta_T &= \frac{\eta_R}{8\mu \max\{1, E_T\}}, \\
u_T^* &= \max\{1, \lceil 8s/\delta_T \rceil\}, \quad u_T = 2^{\text{clog}_2 u_T^*}.
\end{aligned}
\tag{53}$$

Finally define

$$\begin{aligned}
L_h &= \max\{b_2, b_T\} + \max\{q_{\text{al}}, u_T\} + 4, \\
H_{\text{max}} &= L_h + 8q_1 + r_{\text{al}} + r_2 + (D + a) + 10, \\
L_2 &= \max\{2, \text{clog}_2(H_{\text{max}} + 2)\}, \\
k_2 &= a + h + 4096L_2^2 + 8, \quad t_2 = 2^{k_2}.
\end{aligned}
\tag{54}$$

The six-number record is

$$\pi(P, h, I) = (a, s, q_2, t_2, q_{\text{al}}, Q_{\text{br}}), \quad (55)$$

and the proof-side record $g(P, h, I)$ uses $(D, b_0, b_+, k_{\text{ref}}, b_T, u_T, \eta_R, \eta_C, \delta_U)$.

6.3 Direct schedule theorem

Theorem 6.1 (Canonical parameter theorem). *Fix a balanced-family provider satisfying the base contract in [Assumption 4.3](#). Assume $\text{Common}(S)$, $m_R, n_R > 0$, $D \geq 3$, $b_+ \geq 1$, $h \geq 5$, and*

$$k_2 + 2 \leq r_2. \quad (56)$$

Then the values in (47)–(55) satisfy $\text{LevelOK}(\mathcal{F}, P, \pi)$.

Proof. The common source facts supply the three positive carrier fields. From $h \geq 5$ we get $a \geq 7$ and $\sigma = \max\{6, h + 2\} \geq 7$, so $a \geq 2$ and $s = 2^\sigma \geq 4$. The width equation for q_2 is its definition, and $t_2 = 2^{k_2}$ is positive. Equation (56) gives $2^{k_2+2} \leq 2^{r_2}$, which is exactly $4t_2 \leq q_2$.

For the only nontrivial finite-map field, fix one guarded-source coordinate. Its slab label is fixed. At most two vertex items and all $s - 3$ blocker items are active in that slab, so the zero-based rank of every active item is strictly below s . Finally, q_{al} and Q_{br} are positive powers of two. These are precisely the fields of LevelOK . The guarded-coordinate range and balanced-family entry range are the two derived facts identified after [Definition 5.2](#). $\square$

What the theorem does not claim. The preceding theorem is structural. It does *not* derive the following five formula-uniform quantitative rows from arithmetic readiness:

$$\begin{aligned} m_{\text{HS}} &\leq h, & \frac{64sq_1}{\zeta} &\leq q_2, & u_T &\leq q_{\text{al}}, \\ a + 2 + \sigma &\leq b_0, & 100(V_0 + B_0) &\leq T_0. \end{aligned} \quad (57)$$

The first row implies $m_{\text{HS}} + 2 \leq \log_2 s$ because $\log_2 s = \sigma \geq h + 2$. Once all five rows hold, the formula-owned lemmas in [Appendix B](#) give the remaining scalar checklist, including the three reference-bin ceiling gates. This separation prevents a formal proof from obtaining a semantic cutoff inequality merely by destructuring a structural readiness record.

6.4 Formula-independent cutoffs

At the root, choose the robustness depth

$$b(n) = \text{clog}_2(n + 2) \quad (58)$$

and the moderate seed exponent

$$\ell(n) = \log_2^\downarrow(\log_2^\downarrow n), \quad h(n) = \max\{5, \ell(n)^4\}. \quad (59)$$

Here $\log_2^\downarrow$ is the integer floor logarithm, with value zero at zero. Thus (59) is a total natural-number definition.

Lemma 6.2 (Elementary subpower comparison). *Fix positive integers A, d, C . There is $U(A, d, C)$ such that, for every $n \geq U(A, d, C)$,*

$$C 2^{A(1+\ell(n))^d} \leq n. \quad (60)$$

Proof. It is enough to prove

$$\text{clog}_2 C + A(1 + \ell(n))^d \leq \log_2^\downarrow n.$$

For fixed A, d, C , the left side is a fixed polynomial in the binary logarithm of $\log_2^\downarrow n$, whereas the right side is $\log_2^\downarrow n$ itself. The elementary induction $B(1 + \log_2^\downarrow x)^d \leq x$ for all sufficiently large integers x proves the claim after substituting $x = \log_2^\downarrow n$ and increasing the cutoff so that $2^{\log_2^\downarrow n} \leq n$ absorbs the ceiling. This proof uses only monotonicity of integer powers and the two defining inequalities for the integer logarithm. $\square$

Lemma 6.3 (Fixed-depth schedule envelope). *Fix $k \geq 1$ and the two external contracts. There are positive integers A_k, d_k and a cutoff U_k^{env} such that, for every valid formula of size $n \geq U_k^{\text{env}}$, every source-list branch, and every level $i < k$, the following statements hold for the actual dependent state:*

- (E1) *arithmetic readiness holds, including $k_2 + 2 \leq r_2$;*
- (E2) *all five inequalities in (57) hold;*
- (E3) *every ceiling logarithm used in the actual carrier calculation, and every schedule width which is itself required as a naming-width bound, is at most*

$$Z_k(n) = 2^{A_k(1+\ell(n))^{d_k}};$$

- (E4) *the same bound holds for every term in the scalar recurrence (163) through level k .*

The constants are independent of the branch and of the source promise.

Proof. We give the induction in the form needed for transcription. At level zero, the full-cube seed has naming width $q_{\text{seed}} = O(b(n))$. The source contract gives

$$\text{clog}_2(v(G) + e(G) + 2) \leq \text{clog}_2(n(\text{clog}_2(n + 2))^{c_{\text{src}}} + 2),$$

which is bounded by $2^{A(1+\ell(n))^d}$ after increasing fixed A, d .

For the successor step, use the following closed list of elementary closure rules. A finite sum, product, maximum, or fixed power of quantities bounded by $2^{A(1+\ell)^d}$ has another bound of the same form after increasing A, d . The ceiling logarithm of such a quantity is polynomial in $1 + \ell$. Every exponentiation in (47)–(54) is applied either to such a ceiling logarithm or to a fixed polynomial in such logarithms. The balanced-family size bound gives

$$\text{clog}_2 N_2 \leq t_2 \text{clog}_2(2 \max\{q_2, 1000q_1 m_R t_2^2, 2\}),$$

so it obeys the same envelope. Finally, the exact carrier equations (155)–(156) reduce the new naming widths to the polynomial $A(z)$ in (161). Thus one induction step preserves the envelope; a fixed number k of steps is absorbed into A_k, d_k . Taking a maximum over the finite source list before using either source promise makes the bound branch-uniform.

It remains to record why the comparisons, rather than merely the sizes, have the stated direction. The hard-seed row follows because $\ell(n)^4$ tends to infinity. The Stage-2 fit and input-depth row reduce to a fixed polynomial in $1 + \ell(n)$ plus two being at most $\log_2^\downarrow n$; the same comparison, together with $n \leq e(G)$, gives the protected source-width row. The terminal definitions first bound the three projection exponents and e_{max} by the preceding envelope; substituting those bounds in (53) makes the defining alignment exponent dominate $\text{clog}_2 u_T$, which gives $u_T \leq q_{\text{al}}$. For template dominance,

apply (26) on a set of t_2 coordinates. Its strict lower count makes every one of the $(q_1 m_R)^{t_2}$ patterns occur, so $N_2 \geq (q_1 m_R)^{t_2}$. The Stage-2 fit leaves t_2 legal, and the same double-log comparison makes this lower bound exceed $100(V_0 + B_0)/s$. These are precisely the five rows; all other scalar gates are formula consequences listed in Appendix B. $\square$

Lemma 6.4 (Simultaneous finite cutoff). *For every fixed $k \geq 1$ and fixed external source and balanced-family contracts, there is $U_k \geq \max\{k, n_{\text{src}}\}$ such that, for every valid Φ with $n(\Phi) \geq U_k$, every source branch, and every level $i < k$:*

- (C1) *the actual carriers at level i are positive;*
- (C2) *(56) holds for the actual state;*
- (C3) *Theorem 6.1 applies;*
- (C4) *all five rows in (57), and hence the complete scalar checklist and the three reference-bin ceiling gates, hold;*
- (C5) *the computed k -step naming-width majorant is at most $e(n(\Phi), k)$.*

The same U_k works for all branches.

Proof. Apply Lemma 6.3 and then Lemma 6.2 with $A = A_k$, $d = d_k$, and $C = 2k$. After increasing the cutoff, every envelope term is at most $\lfloor n/(2k) \rfloor = e(n, k)$. Take U_k to be the maximum of the envelope cutoff, the subpower cutoff, k , and n_{src} . Conditions (C1)–(C4) are exactly (E1)–(E2) plus the structural and formula-owned consequences already proved above. Condition (C5) is (E4) followed by the subpower comparison. The construction of U_k uses only uniform upper and lower envelopes; it assumes no monotonicity of the actual source list as the formula size changes. Appendix B gives the denominator-cleared conversion used after (C5). $\square$

7 The total one-level construction

Fix public input $P = (S, m_R, n_R, R)$ and parameters $\pi = (a, s, q_2, t_2, q_{\text{al}}, Q_{\text{br}})$. The balanced family is $\mathcal{F}(q_2, t_2, q_1 m_R) = (f_\ell)_{\ell \in [N_2]}$. Put

$$q_1 = 2^a - 2, \quad L_1 = q_1 + s + 1.$$

This section defines every matrix before either source predicate is used.

7.1 Stages 1–3

The first stage is

$$\begin{aligned} X_1 &= [q_1] \times [m_R], & Y_1 &= ([L_1] \rightarrow [n_R]), \\ M_1((i, x), \Gamma) &= R(x, \Gamma(i)). \end{aligned} \tag{61}$$

The initial inclusion $[q_1] \hookrightarrow [L_1]$ is $i \mapsto i$. The tail inclusion $[s+1] \hookrightarrow [L_1]$ is $r \mapsto q_1 + r$.

Identify $[q_1 m_R]$ with $[q_1] \times [m_R]$ by the fixed quotient/remainder bijection. For $\ell \in [N_2]$ and $\alpha \in [q_2]$, let

$$c_{\ell, \alpha} \in X_1$$

be the image of $f_\ell(\alpha)$. Its existence as an element of X_1 is exactly the balanced-family range assertion.

The second stage has

$$\begin{aligned} X_2 &= [q_2] \times Y_1, & Y_2 &= [N_2], \\ M_2((\alpha, \Gamma), \ell) &= M_1(c_{\ell, \alpha}, \Gamma). \end{aligned} \quad (62)$$

The third stage is the ordinary s -copy interlace of $M_2^\top$:

$$\begin{aligned} X_3 &= [s] \times Y_2, & Y_3 &= ([s] \rightarrow X_2), \\ M_3((p, \ell), d) &= M_2(d(p), \ell). \end{aligned} \quad (63)$$

7.2 Guarded source rows

Let $V = [v(G)]$ and let $E = [e(G)]$. An incidence position is a pair $(e, b) \in E \times [2]$, whose vertex is $\partial_b e$. Define

$$\begin{aligned} \mathcal{I} &= V \sqcup [s-3], & \mathcal{Z} &= E \sqcup (E \times [2]), \\ X_{\text{src}} &= [4s] \times \mathcal{I}, & Z_{\text{real}} &= [4s] \times \mathcal{Z}. \end{aligned} \quad (64)$$

For $u \in V$, an edge coordinate accepts u precisely when u is an endpoint; an incidence coordinate accepts u precisely when it names u . Every blocker item in $[s-3]$ accepts every local coordinate. Denote this Boolean relation by $\text{local}(i, z)$.

The zero-based ranks are

$$\begin{aligned} \text{rank}_{\mathcal{I}}(u) &= u & (u \in V), \\ \text{rank}_{\mathcal{I}}(r) &= v(G) + r & (r \in [s-3]), \\ \text{rank}_{\mathcal{Z}}(e) &= e & (e \in E), \\ \text{rank}_{\mathcal{Z}}(e, b) &= e(G) + 2e + b & ((e, b) \in E \times [2]). \end{aligned}$$

For $(h, i) \in X_{\text{src}}$ and $(h', z) \in Z_{\text{real}}$, the slab labels must agree. The resulting guarded source matrix $G_{\text{src}} : X_{\text{src}} \times [q_2] \rightarrow \{0, 1\}$ is

$$G_{\text{src}}((h, i), \alpha) = 1 \iff \exists z \in \mathcal{Z} : \alpha = h(3e(G)) + \text{rank}_{\mathcal{Z}}(z) \wedge \text{local}(i, z) = 1. \quad (65)$$

The equation is interpreted in $[q_2]$ using the rank bound from **LevelOK**. For each α , order the active source rows by their ordinary rank in X_{src} ; the active-rank theorem assigns their ranks to $[s]$. This deterministic order is the only tie-breaking rule.

7.3 Raw Stage 4

Let a seed tuple be $\beta : [s+1] \rightarrow [n_R]$. Define

$$\begin{aligned} X_B &= ([s] \times Y_2) \sqcup (X_{\text{src}} \times [m_R]), \\ Y_B &= ([s+1] \rightarrow [n_R]) \times ([s] \rightarrow X_2). \end{aligned} \quad (66)$$

For $c = (\beta, d) \in Y_B$, let $\alpha(c)$ be the Stage-2 coordinate in $d(0)$. The structural checklist identifies it with a guarded-source coordinate. The raw matrix $B : X_B \times Y_B \rightarrow \{0, 1\}$ is

$$\begin{aligned} B((p, \ell), c) &= M_2(d(p), \ell), & (67) \\ B(((h, i), x), c) &= \begin{cases} R(x, \beta(r_{\alpha(c)}(h, i))), & \text{all } d(p) \text{ advertise } \alpha(c), \\ R(x, \beta(s)), & \text{otherwise,} \end{cases} & \begin{matrix} G_{\text{src}}((h, i), \alpha(c)) = 1, \\ \end{matrix} & (68) \end{aligned}$$

where $r_{\alpha}(h, i) \in [s]$ is the active rank. Thus the neutral branch uses the last seed-tuple coordinate, while active rows use one of the first s . Every symbol in (67)–(68) was fixed before the promise.

7.4 Protected labelled duplication

Four row types and three column types are duplicated. Let

$$\begin{aligned} T &= [s] \times Y_2, \\ V_S &= ([4s] \times V) \times [m_R], \\ B_S &= ([4s] \times [s-3]) \times [m_R], \\ S_C &= ([s] \rightarrow X_2), \\ D_C &= [q_2] \times Y_1. \end{aligned}$$

These carriers have fixed inclusions into X_B and Y_B : T, V_S, B_S use the corresponding summands of X_B ; a selector $d \in S_C$ is paired with the constant seed tuple at column 0; and $(\alpha, \Gamma) \in D_C$ is sent to the column whose seed tuple is the tail of Γ and whose selector is constantly (α, Γ) .

Write

$$\begin{aligned} M_0 &= |Y_B|, & S_0 &= |S_C|, & D'_0 &= |D_C|, \\ X_0 &= |X_B|, & T_0 &= |T|, & V_0 &= |V_S|, & B_0 &= |B_S|. \end{aligned}$$

The column multiplicities are

$$w_{\text{base}} = 182S_0D'_0, \quad w_{\text{sel}} = M_0D'_0, \quad w_{\text{diag}} = M_0S_0, \quad (69)$$

and the row multiplicities are

$$\begin{aligned} u_{\text{base}} &= (2^{22} - 4)T_0V_0B_0, & u_{\text{temp}} &= 2X_0V_0B_0, \\ u_{\text{vert}} &= X_0T_0B_0, & u_{\text{block}} &= X_0T_0V_0. \end{aligned} \quad (70)$$

The protected row and column carriers are the labelled disjoint unions

$$X_A = (X_B \times [u_{\text{base}}]) \sqcup (T \times [u_{\text{temp}}]) \sqcup (V_S \times [u_{\text{vert}}]) \sqcup (B_S \times [u_{\text{block}}]), \quad (71)$$

$$Y_A = (Y_B \times [w_{\text{base}}]) \sqcup (S_C \times [w_{\text{sel}}]) \sqcup (D_C \times [w_{\text{diag}}]). \quad (72)$$

Let $\rho : X_A \rightarrow X_B$ and $\kappa : Y_A \rightarrow Y_B$ forget labels and apply the displayed inclusions. Then

$$A(r, c) = B(\rho(r), \kappa(c)). \quad (73)$$

The weights are positive on every call used later. They give exactly base, template, vertex-source, and blocker-source row reservoirs and base, selector, and diagonal column reservoirs. Equal labels produce equal rows or columns, so [Lemma 3.2\(d\)](#) applies.

7.5 Forced post-composition

Set

$$N = A^\top. \quad (74)$$

Define the reference row carrier and inclusion literally by

$$X_{N_{\text{ref}}} = Y_B \times [w_{\text{base}}], \quad \iota_{\text{ref}}(c, \lambda) = \text{inl}(c, \lambda) \in Y_A = \text{Rows}(N). \quad (75)$$

Thus the reference rows are all base copies of all raw columns, with exactly the base multiplicity w_{base} ; the inclusion is the first-summand map and is injective. The column carrier is all of $X_A = \text{Cols}(N)$, and

$$N_{\text{ref}}((c, \lambda), r) = N(\iota_{\text{ref}}(c, \lambda), r). \quad (76)$$

Finally define U, W, R^+ by (12)–(13) and let

$$\text{Succ}(P, \pi) = \text{Can}(R^+) \quad (77)$$

be the canonical zero-based reindexing. If α and β are the canonical carrier bijections, then

$$\text{Succ}(P, \pi)(i, j) = R^+(\alpha(i), \beta(j)).$$

There is no other public successor.

Theorem 7.1 (Totality and genuine-branch equations). *The definitions $T, B, A, N, N_{\text{ref}}, U, W, R^+$ and Succ are total functions of P, π and balanced-family data. If $\text{LevelOK}(\mathcal{F}, P, \pi)$ holds, their values are exactly (61)–(77); otherwise every guarded core object uses its fixed all-false branch on the same displayed carriers. On the genuine branch the carrier equations (14)–(15) hold literally.*

Proof. Decide the structural proposition once. In the positive branch every bounded index and inclusion is supplied by the corresponding checklist field. In the negative branch the all-false functions require no element of any carrier. The post-composition is unconditional. The cardinalities follow from the standard formulas for finite sums, products, and function spaces. Canonical reindexing preserves them by bijection. $\square$

8 Protected Stage 4 on the actual matrix

Fix P, π, g satisfying LevelOK and NoLevelOK . All matrices in Section 7 are already fixed. Put $C = C(P, \pi, g)$ as in (32).

8.1 The exact protected statement

Theorem 8.1 (Protected fourth-stage certificate). *For the actual matrices N and N_{ref} defined in (74)–(76):*

$$C \leq \mathcal{D}_N(1; 1 - \eta_R, 1 - \eta_C), \quad (78)$$

$$D(N) = C, \quad (79)$$

$$|\text{Cols}(N_{\text{ref}})| = |\text{Cols}(N)|, \quad (80)$$

$$C - 1 - r \leq \mathcal{D}_{N_{\text{ref}}}(1; 2^{-b_T}, (3/5)/2^r) \quad (r = 0, 1, 2). \quad (81)$$

The matrices and the reference inclusion are the same objects constructed before the negative source predicate was invoked.

We prove the four claims in dependency order. The proof never requests a low-density full-cost estimate for all of N ; the counterexamples described in Section 1 show why that stronger target would be false.

8.2 Dense retained members

Fix a retained pair (R', C') for N with

$$|R'| \geq \lceil (1 - \eta_R) |\text{Rows}(N)| \rceil, \quad |C'| \geq \lceil (1 - \eta_C) |\text{Cols}(N)| \rceil,$$

and fix a protocol $\mathcal{P}$ computing $N|_{R' \times C'}$. Suppose for a contradiction that $\text{cost}(\mathcal{P}) < C$.

The duplication weights let us count damage in each labelled reservoir rather than in the raw matrix. A base object is called *good* when at least the prescribed fraction of its labels survives. Markov's inequality in cleared integer form gives the following consequences of (41):

- (P1) at least a $1 - 1/1000$ fraction of template rows have enough surviving labels;
- (P2) all but a $\delta_V/16$ fraction of vertex-source rows have enough surviving labels;
- (P3) all but a $1/(16(s-3))$ fraction of blocker-source rows have enough surviving labels;
- (P4) every needed selector and diagonal column class has a surviving representative after the row loss η_R in N (equivalently, the loss of protected columns in A).

The denominators in these statements are positive by LevelOK.

For each source coordinate $\alpha \in [q_2]$, the surviving template rows determine the Stage-2 patterns visible at that coordinate. If too many coordinates fail to contain the required pattern, the balanced count (26) and $8q_1 \leq t_2$ give more missing family members than the total column loss permits. Hence all but an explicitly bounded set $E_0 \subseteq [q_2]$ are simultaneously good for the singleton, pair, and reference projections used below. The projection-cover calculation is finite: for a product family at density ρ , the least cover exponent B satisfies $(1 - \rho)^B < \theta$; the union of the three covers has size at most the number $e_{\max}$ in (53).

8.3 Source synchronization

Follow $\mathcal{P}$ through Alice queries while keeping every surviving template row compatible with the current transcript. A coordinate is *represented* at a node if its surviving template rows still display the Stage-2 pattern needed to read that coordinate. The following invariant is initialized at the root and preserved by every Alice query.

Invariant 8.2 (Protected routing invariant). At a reached node v there are:

- (I1) a nonempty compatible row set and its exact remaining protocol;
- (I2) a set $J_v \subseteq [q_2] \setminus E_0$ of represented coordinates;
- (I3) for every $\alpha \in J_v$, a surviving template block and the corresponding selector and diagonal column representatives;
- (I4) a partial source colouring determined by the answers already fixed;
- (I5) a nonnegative integer debt d_v equal to the number of lower profile rungs already spent;
- (I6) the inequality

$$\text{cost}(\mathcal{P}_v) < C - |\text{route}(v)|,$$

where the natural subtraction is defined because $|\text{route}(v)| < C$, together with the exact row-mass lower bound needed by the next extension call.

Every stored row and column is an element of the literal retained submatrix; no abstract replacement game occurs.

At an Alice node, partition the compatible rows by her answer. Assign each represented coordinate to a child in which its certified template block survives. Conservation of row labels and the defect bounds (38)–(40) show that the sum of the two children’s certified masses is at least the parent’s mass. Choose a nonempty child meeting the displayed lower bound. At a Bob node, the columns split. The selector or diagonal representative for a coordinate either remains in a child or pays one unit of debt, and the three-rung input profile of R supplies the corresponding lower bound. These are the only two local transition types.

If synchronization failed for every represented coordinate, the source-row answers would define a three-colouring of G with fewer than $\gamma e(G)$ monochromatic occurrences. The vertex-source rows force consistency at incidences, and the blocker rows force the unused colours to remain distinct. The loss bounds in (P1)–(P3), together with $64sq_1/\zeta \leq q_2$, ensure that fewer than a γ fraction of edge occurrences can be discarded. This contradicts $\text{SourceNo}(S)$.

8.4 Stopping frontier and contradiction

Stop a routed branch at the first of two events: Bob speaks, or every selector bit required by its surviving coordinate has been exposed. The reached stopping nodes form a prefix-free frontier. The row sets reaching distinct nodes are disjoint, and every retained row reaches exactly one node. Moreover, a binary node at route length d has capacity 2^{-d} ; Kraft’s inequality bounds the total capacity of the prefix-free frontier by one.

The weighted frontier lemma applies these two conservation laws to [Invariant 8.2](#). It returns one stopping node with enough ordinary template mass and enough source-owned mass for a concrete local extension. There are two cases.

- (F1) If every selector bit has been exposed, the surviving selector and diagonal columns embed a restriction of the incoming seed matrix. Its appropriate rung in (19), plus the selector bits already paid, gives remaining cost at least $C - |\text{route}(v)|$.
- (F2) If Bob speaks first, one of his actual children contains the two required source/template groups and paired columns. The fixed protected extension statement (call (i) in [Appendix A](#)) gives the same lower bound in that child.

Either conclusion contradicts the strict cost assumption on $\mathcal{P}$. Thus every protocol for every retained member costs at least C , proving (78).

8.5 Exact cost

Take the full row and column sets in (78). They are a member of the bracket family because $0 < 1 - \eta_R, 1 - \eta_C \leq 1$. Hence $D(N) \geq C$.

For the reverse inequality, first run the explicit raw fourth-stage protocol. Alice sends the a Stage–1 selector bits; the parties then send the $\log_2 q_2$ Stage–2 coordinate bits and the $\log_2 s$ Stage–3 selector bits. One final bit distinguishes the template and guarded-source cases, and the remaining conversation is an optimal protocol for R . Conditions (37) ensure that every selector is in range. Thus

$$D(B) \leq D_0 + a + \log_2 q_2 + \log_2 s + 1 = C.$$

Duplicating labelled rows and columns preserves cost, and transposition preserves cost, so $D(N) \leq C$. This proves (79). Notice that the upper protocol is explicit; exact cost is not inferred from the lower bound alone.

8.6 The actual reference restriction

The reference carrier is all base-column copies of all raw columns, so (80) follows directly from (76): only rows of N are restricted.

Fix $r \in \{0, 1, 2\}$ and a retained member at row density 2^{-br} and column density $(3/5)/2^r$. The template-density formula and the reference ceiling inequality in Definition 5.4 select at least $9s/(16 \cdot 2^r)$ actual Stage-3 blocks. The physical reference landing map sends their rows and all retained columns into the literal member of N_{ref} ; it is injective on rows and preserves every entry. The fixed reference extension theorem then embeds the appropriate rung of the incoming seed. Its cost is at least $C - 1 - r$. Restriction monotonicity yields (81).

This completes Theorem 8.1.

9 Scale alignment on the actual retained game

The protected theorem supplies a full-cost estimate near the whole of N and three reference estimates at much smaller row density. This section aligns them. Put

$$U = \text{Int}_{q_{\text{al}}}(N), \quad W = U^{\top}, \quad C_W = C + \log_2 q_{\text{al}}, \quad \theta = \frac{1}{160Q_{\text{br}}}.$$

9.1 Operational target

Theorem 9.1 (No shallow protocol for the actual alignment game). *For every retained pair (R_U, C_U) satisfying*

$$|R_U| \geq \lceil (1 - \delta_U) |\text{Rows}(U)| \rceil, \quad |C_U| \geq \lceil \theta |\text{Cols}(U)| \rceil, \quad (82)$$

and every protocol Q computing $U|_{R_U \times C_U}$,

$$\text{cost}(Q) \geq C_W. \quad (83)$$

This statement concerns the literal U in (12). It does not quantify over an arbitrary trace or an arbitrary matrix with similar cardinalities.

9.2 Physical coordinates and reference rows

A row of U is (i, r) with $i \in [q_{\text{al}}]$ and $r \in \text{Rows}(N)$; a column is a function $f : [q_{\text{al}}] \rightarrow \text{Cols}(N)$; and

$$U((i, r), f) = N(r, f(i)).$$

For each coordinate i , define the retained physical row fibre

$$R_i = \{r : (i, r) \in R_U\}.$$

Let $\iota_{\text{ref}} : \text{Rows}(N_{\text{ref}}) \hookrightarrow \text{Rows}(N)$ be the literal inclusion from (76), and define the retained reference fibre by

$$R_i^{\text{ref}} = \{r_{\text{ref}} \in \text{Rows}(N_{\text{ref}}) : (i, \iota_{\text{ref}}(r_{\text{ref}})) \in R_U\}.$$

Thus R_i^{ref} is an intersection with the actual retained row set; it is not a fresh copy of the reference carrier. Summing the deleted rows over coordinates and using (82) gives

$$\sum_{i < q_{\text{al}}} (|\text{Rows}(N)| - |R_i|) \leq \delta_U q_{\text{al}} |\text{Rows}(N)| = \frac{\eta_R}{16} |\text{Rows}(N)|. \quad (84)$$

The same sum bounds the aggregate deficit of the reference fibres, because each missing reference row is the image of a missing physical row. The clean-coordinate counting lemma, applied to this aggregate bound, proves that all but the scheduled number of coordinates have the reference size needed later. In particular, the proof never assumes separately that every coordinate retains enough reference rows.

The column projection theorem applied to C_U constructs exceptional sets for the singleton, paired, and small-block projections used below. Enlarge their union by the row-bad coordinates from the preceding paragraph and call the resulting set $E_0 \subseteq [q_{\text{al}}]$. The clean-coordinate and product-cover counts give $|E_0| \leq e_{\text{max}}$. Every coordinate outside E_0 therefore has both the initial reference-row floor and all required column projections. The projection theorem is elementary: if every B -tuple projection missed one pattern, at most a $(1 - \rho)^B$ fraction of the full product could remain; the definition of B makes this smaller than θ .

9.3 The concrete protocol frontier

Assume for a contradiction that Q in [Theorem 9.1](#) has cost at most $C_W - 1$. Write $r_{\text{al}} = \log_2 q_{\text{al}}$. Starting with all rows in R_U , follow only actual Alice questions of Q . Stop either at the first Bob node, necessarily before depth r_{al} , or after r_{al} Alice answers. If an Alice-only branch reaches a leaf earlier, attach a fixed virtual bit string to its route until the route has length r_{al} . This string is bookkeeping only: it creates no question, changes no output, and leaves the remaining protocol at that address equal to the original leaf. The first-Bob routes and the length- r_{al} final route tokens form a prefix-free set $\mathcal{T}$.

For $v \in \mathcal{T}$, let $\mathcal{R}_v$ be the rows assigned to its route token, let d_v be the token length, and let Q_v be the literal remaining subtree of Q at the last actual node on that token. Define the normalized capacity $\text{cap}(v) = 2^{-d_v}$ and the residual dyadic capacity

$$u_v = 2^{r_{\text{al}} - d_v}.$$

Thus $u_v \geq 2$ at a first-Bob node and $u_v = 1$ at a final token. Direct induction on the actual Alice queries, followed by the harmless virtual extension of early leaves, gives

$$R_U = \bigsqcup_{v \in \mathcal{T}} \mathcal{R}_v, \tag{85}$$

$$\sum_{v \in \mathcal{T}} \text{cap}(v) \leq 1, \tag{86}$$

$$\text{cost}(Q_v) \leq C_W - 1 - |\text{route}(v)|. \tag{87}$$

In the last display $|\text{route}(v)| = d_v$, including virtual bits at an early leaf. Indeed the leaf suffix has cost zero, so its inequality follows from $C \geq 1$. Empty route classes are discarded before taking the frontier.

The reference rows are routed by certificates, not by pretending that all rows in one physical coordinate follow one branch. Fix the reference base floor A_0 from the schedule and put

$$A_k = 2^k A_0 \quad (0 \leq k \leq r_{\text{al}}).$$

At the root, every coordinate outside E_0 starts with an actual subset of R_i^{ref} of cardinality at least $A_{r_{\text{al}}}$. At an Alice node of depth d , intersect this subset with the two real row children. If one child contains at least $A_{r_{\text{al}} - d - 1}$ reference rows, it carries a descendant certificate. If both children meet the threshold, record two split descendants; otherwise the qualifying child is the unique clean

descendant. This is exhaustive because $A_{r_{\text{al}}-d} = 2A_{r_{\text{al}}-d-1}$. Stop a descendant at the first Bob node or at a final route token.

Let $\mathcal{I}$ be the resulting finite certificate index set. Every $j \in \mathcal{I}$ stores its physical coordinate $\text{coord}(j)$, its stopping token $\text{place}(j)$, its actual reference-row subset, and the tag clean or split. A fixed token receives at most one descendant of a fixed starting certificate. Independently, the protected rows in physical coordinate i are partitioned by their actual Alice answers and may reach several stopping tokens. This distinction between a certificate route and physical protected-row mass is used throughout the owner–donor count.

Invariant 9.2 (Alignment frontier invariant). For every reached node v :

- (A1) Q_v computes the literal submatrix on the rows and columns reaching v ;
- (A2) the row sets, reference fibres, and coordinate projections are exact subsets of R_U and C_U ;
- (A3) every routed certificate has one stored coordinate, one stored token, one clean-or-split tag, and one residual dyadic capacity u_v ;
- (A4) physical row mass is partitioned at Alice nodes; at a Bob node the row set is unchanged while the columns split between the two real children;
- (A5) the total exceptional and early-loss charges are bounded by $e_{\max}$ and (84);
- (A6) the remaining cost is the exact suffix bound (87).

Initialization is substitution at the root. Alice preservation partitions rows by her actual query and applies the preceding certificate rule. Bob preservation leaves the rows fixed, partitions columns by his actual query, and examines the real child selected by the counting inequality. No extra question or modified protocol is introduced.

9.4 Hard first-Bob nodes

Suppose v is a first-Bob node. If its residual dyadic capacity is small, the actual routed reference rows and the projection cover embed a member of the reference restriction N_{ref} . The concrete small-capacity lemma selects one of the three bounds in (81) and proves

$$\text{cost}(Q_v) \geq C_W - d_v,$$

exactly one more than the upper bound in (87).

If the capacity is large, select a dyadic class of donor coordinates. The first product clause (B1) in [Assumption 4.3](#) gives the precise balancedness of the selected product family. The concrete extension call (iii) from [Appendix A](#) produces one real Alice child inside a real Bob child with the full Stage–2 and Stage–3 certificate. The ordinary and foreign row fibres and the paired columns are embedded by the literal carrier maps. The heavy-node arithmetic and the fixed extension call prove for this literal suffix the same lower bound $\text{cost}(Q_v) \geq C_W - d_v$, again contradicting (87).

Therefore an alleged shallow protocol can place only a bounded amount of protected mass from clean certificates at first-Bob nodes. The precise count is the E_B term used below.

9.5 Final owners, donors, and global mass

We now give the complete terminal count. Put $\vartheta = 9/16$. Let $J_{\text{clean}} \subseteq \mathcal{I}$ be the clean certificates ending at final Alice tokens, let $c = |J_{\text{clean}}|$, let n_T be the number of final tokens, and let

$$K_B = \sum_{v: \text{first Bob}} u_v.$$

Certificate conservation, the first-Bob light estimate, and Kraft's inequality give

$$c \geq q_{\text{al}} - \frac{e_{\text{max}}}{1 - \vartheta}, \quad K_B + n_T \leq q_{\text{al}}. \quad (88)$$

For a first-Bob token v and a set $J \subseteq J_{\text{clean}}$, let $P_v(J)$ be the protected row mass from the physical coordinates of J which reaches v , divided by $m = |\text{Rows}(N)|$. The contrapositive of the hard large-capacity conclusion gives

$$P_B := \sum_{v: \text{first Bob}} P_v(J_{\text{clean}}) < 128\mu B_{\text{pair}} L_q s K_B \leq \frac{128\mu B_{\text{pair}} L_q s e_{\text{max}}}{1 - \vartheta}. \quad (89)$$

Call $j \in J_{\text{clean}}$ *Bob-heavy* when $\sum_v P_v(\{j\}) > \eta_R/4$. Since all summands are nonnegative, (89) and Markov counting give

$$e_B := |J_{\text{heavy}}| \leq \frac{512\mu B_{\text{pair}} L_q s e_{\text{max}}}{(1 - \vartheta)\eta_R} \leq E_B. \quad (90)$$

Define the good donors by $J_{\text{good}} = J_{\text{clean}} \setminus J_{\text{heavy}}$.

For $j \in J_{\text{clean}}$, its *owner* $o(j)$ is the final token at which its clean certificate stops. The final-reference uniqueness lemma says that a final token contains at most one clean certificate. Hence o is injective on J_{clean} . The *exceptional final tokens* are

$$\mathcal{E}_T = \{v : v \text{ is final}\} \setminus o(J_{\text{good}}), \quad e_T = |\mathcal{E}_T|.$$

Using (88) and $|J_{\text{good}}| = c - e_B$ gives the literal cardinality estimate

$$e_T = n_T - |J_{\text{good}}| \leq q_{\text{al}} - c + e_B \leq \left\lceil \frac{e_{\text{max}}}{1 - \vartheta} \right\rceil + E_B = E_T. \quad (91)$$

The first terminal wrapper gate in (179) gives $E_T \leq q_{\text{al}}/8$. The definition of E_T gives $e_{\text{max}}/(1 - \vartheta) + E_B \leq E_T$. Therefore (88) and (90) give

$$|J_{\text{good}}| = c - e_B \geq q_{\text{al}} - \frac{e_{\text{max}}}{1 - \vartheta} - E_B \geq q_{\text{al}} - E_T \geq \frac{7}{8}q_{\text{al}} > \frac{1}{2}q_{\text{al}}.$$

In particular

$$J_{\text{good}} \neq \emptyset. \quad (92)$$

For a donor j and a final token T , let $m_{j,T}$ be the protected mass from the physical coordinate $\text{coord}(j)$ which reaches T , again normalized by m . Four disjoint sets account for protected rows of a good donor missing from its owner:

initially deleted rows	at most $\eta_R m/16$
ordinary rows lost from the unique clean certificate route	less than $\eta_R m/16$
protected rows stopped at first-Bob tokens	at most $\eta_R m/4$
protected rows reaching other final tokens	the remaining loss

For the second row, a clean certificate at remaining capacity 2^k loses fewer than A_{k-1} reference rows at the unchosen Alice child. Hence its total route loss is less than $\sum_{h < r_{\text{al}}} A_h = (q_{\text{al}} - 1)A_0 < q_{\text{al}}A_0$, and the schedule bounds the latter by $\eta_R m/16$. The third row is exactly the negation of Bob-heaviness.

If the last row had mass at most $\eta_R m/2$, the owner would retain more than

$$1 - \left(\frac{1}{16} + \frac{1}{16} + \frac{1}{4} + \frac{1}{2} \right) \eta_R = 1 - \frac{7}{8} \eta_R > 1 - \eta_R$$

of its full physical copy of N . Its coordinate lies outside the singleton projection cover, so its retained outer-column projection has density at least $1 - \eta_C$. The protected Stage-4 theorem would then force residual cost at least C at the final token, whereas (87) gives at most $C_W - 1 - r_{\text{al}} = C - 1$. Therefore

$$\sum_{T \neq o(j)} m_{j,T} > \eta_R/2 \quad (j \in J_{\text{good}}). \quad (93)$$

It remains to show that little of this outgoing mass can reach owners of other good donors. Let m_G be the ordinary baseline size in one physical copy and put

$$g = \frac{m_G}{m} = \frac{91}{92}.$$

For $i \in J_{\text{good}}$, let ε_i be the fraction of this ordinary baseline missing at $o(i)$, and define

$$\lambda_0 = q_2 2^{1-b_1}, \quad \delta_i = \lambda_0 + 5\varepsilon_i + \frac{2}{q_2}. \quad (94)$$

Disjoint owner baselines charge every initial deletion once, and every clean route loses less than $(q_{\text{al}} - 1)A_0$ reference rows. Consequently

$$\sum_{i \in J_{\text{good}}} \varepsilon_i \leq \frac{\eta_R}{8g} < \frac{1}{5}. \quad (95)$$

Fix distinct $j, i \in J_{\text{good}}$. If

$$m_{j,o(i)} \geq 2\mu\delta_i, \quad (96)$$

then one of the two physical protected reservoirs of donor j has relative density at least δ_i at $o(i)$. The ordinary rows of i have deficit ε_i there. Clause (B2), the literal Stage-4 carrier maps, and extension call (ii) therefore construct the actual two-stream terminal restriction at $o(i)$ and give it communication cost at least C . But $o(i)$ is final, so its literal suffix costs at most $C - 1$. Thus (96) is impossible, and

$$m_{j,o(i)} < 2\mu\delta_i \quad (i \neq j \text{ in } J_{\text{good}}). \quad (97)$$

This is the pairwise terminal statement: it is a fresh restriction for the ordered pair (j, i) , not a reusable conclusion-bearing record.

Distinct good donors have distinct owners, so for a fixed j the left side below is exactly the mass sent by j to other good owners. The three owner gates in Appendix B and (95) give

$$\begin{aligned} \sum_{\substack{i \in J_{\text{good}} \\ i \neq j}} m_{j,o(i)} &< 2\mu \sum_{i \in J_{\text{good}}} \delta_i \\ &\leq 2\mu \left(q_{\text{al}}\lambda_0 + \frac{5\eta_R}{8g} + \frac{2q_{\text{al}}}{q_2} \right) \leq \frac{3\eta_R}{16} < \frac{\eta_R}{4}. \end{aligned} \quad (98)$$

Combining (93) and (98), every good donor sends more than $\eta_R/4$ protected mass to exceptional final tokens. If $e_T = 0$, this is already a contradiction.

Assume $e_T > 0$. For each good donor, choose an exceptional token receiving more than $\eta_R/(4e_T)$ of its mass. Some $T_* \in \mathcal{E}_T$ is chosen by at least $|J_{\text{good}}|/e_T$ donors. Splitting their mass between the two physical protected reservoirs and choosing the more frequent reservoir type leaves a donor set J_* with

$$|J_{\text{good}}| \leq 2e_T|J_*|, \quad \delta_T \leq \frac{\eta_R}{8\mu e_T} \leq \text{density}_{T_*}(j) \quad \text{for every } j \in J_*. \quad (99)$$

Here the first inequality in the density line uses $e_T \leq E_T$ and the definition of δ_T .

The terminal wrapper gates state

$$4E_T u_T B_{\text{pair}} < q_{\text{al}}, \quad u_T \delta_T \geq 8s. \quad (100)$$

Together with $|J_{\text{good}}| > q_{\text{al}}/2$ and (99), the first gate leaves at least $u_T B_{\text{pair}}$ donors at T_* . The projection-cover theorem selects u_T of them whose joint outer projection is rich. Fix a half-block $J' \subseteq [u_T]$ of size $u_T/2$. Clause (B3) is exactly the balancedness of the literal product on $J' \times [q_2]$.

For every selected donor, partition its protected rows at T_* by Stage-2 coordinate. Averaging and the second gate in (100) provide at least $2sq_2$ represented Stage-2 blocks, each with the required row-density floor. Select the literal number demanded by extension call (iv). The call constructs a restriction of the actual suffix at T_* , using only rows already present there and the actual localized product columns, and proves that its cost is at least C . Since T_* is final, the same suffix has cost at most $C - 1$. This contradiction completes the proof of [Theorem 9.1](#).

9.6 Public scale-aligned certificate

Theorem 9.3 (Scale-aligned certificate). *Under the hypotheses of [Theorem 8.1](#), the actual matrix W satisfies*

$$D(W) = C_W, \quad (101)$$

$$C_W \leq \mathcal{D}_W(1; \theta, 1 - \delta_U), \quad (102)$$

$$C_W - 1 \leq \Lambda_W(1; \theta, 3/5). \quad (103)$$

If $D(N) \leq Y_N$, then

$$D(W) \leq Y_N + \log_2 q_{\text{al}}. \quad (104)$$

Proof. The bracket family in [Theorem 9.1](#) is nonempty because its two densities lie in $(0, 1]$ and the carriers are positive. Minimizing (83) gives $C_W \leq \mathcal{D}_U(1; 1 - \delta_U, \theta)$. At one copy, transposition swaps the two retained densities, proving (102). Taking the full member gives $D(W) \geq C_W$, while the selector protocol (11) and (79) give the reverse inequality and hence (101).

For (103), apply the three actual reference bounds (81) coordinatewise inside U . The reference width, row floor, and wrapper-width conditions in (43) select enough coordinates at each of the three column densities. Transpose and substitute the definition (17). Finally, the same selector protocol applied to an arbitrary upper bound on N gives (104). $\square$

10 The construction-independent bridge

This section is independent of the fourth-stage carrier definitions. It takes a finite matrix with the three common-threshold facts proved in [Theorem 9.3](#) and returns an exact robust successor. Every projection below is a projection of an actual retained row or column set.

Definition 10.1 (Common-threshold input). For a nonempty finite matrix W , an integer $C_W \geq 1$, and reals $0 < \theta$ and $0 < \delta < 1$, write $\text{CommonThreshold}(W, C_W, \theta, \delta)$ for

$$D(W) = C_W, \quad (105)$$

$$C_W \leq \mathcal{D}_W(1; \theta, 1 - \delta), \quad (106)$$

$$C_W - 1 \leq \Lambda_W(1; \theta, 3/5). \quad (107)$$

Let $Q = \text{Qwidth}(b, \delta)$ be the least positive power of two satisfying

$$80b \leq Q\delta, \quad (108)$$

put $r_Q = \log_2 Q$, and define

$$\text{Bridge}(W; b, \delta) = \text{Int}_Q(W)^\top. \quad (109)$$

Theorem 10.2 (Rational common-threshold bridge). *Let W have nonempty finite carriers, let $C_W \geq 3$ and $b \geq 1$, and let $0 < \theta$ and $0 < \delta \leq 1/64$. Assume $\text{CommonThreshold}(W, C_W, \theta, \delta)$ and*

$$Q\theta < \frac{1}{20}. \quad (110)$$

For $R^+ = \text{Bridge}(W; b, \delta)$,

$$D(R^+) = C_W + r_Q, \quad (111)$$

$$C_W + r_Q \leq \text{Rob}(R^+; b, 3/5). \quad (112)$$

Moreover, every upper bound $D(W) \leq Y_W$ gives

$$D(R^+) \leq Y_W + r_Q. \quad (113)$$

10.1 The two inherited base estimates

Expanding (107) gives three one-copy estimates. The finite two-copy ladder applied to those three estimates gives

$$C_W \leq \Lambda_W(2; 2\theta, (3/5)^2). \quad (114)$$

Because $Q \geq 1$, (110) implies $4\theta < 1$. The finite three-copy row-step lemma, applied to (114), therefore gives

$$C_W + 1 \leq \mathcal{D}_W(3; 4\theta, (3/5)^2). \quad (115)$$

These are the only communication lower bounds used by the bridge besides (106). Both are inherited finite lemmas, not new hypotheses.

10.2 Defect bookkeeping

Write $m = |\text{Rows}(W)|$, $n = |\text{Cols}(W)|$, and

$$\lambda = -\ln(1 - \delta). \quad (116)$$

For $a \geq 0$ define the row floor

$$A_a = \lceil m2^a\theta \rceil. \quad (117)$$

Whenever $a \geq 1$,

$$A_{a-1} \leq \lceil A_a/2 \rceil. \quad (118)$$

Indeed $A_a \geq m2^a\theta$, and taking ceilings after division by two gives the displayed inequality.

For a nonempty column set $K \subseteq \text{Cols}(W)^p$, define its logarithmic defect by

$$\text{Def}_p(K) = \ln \frac{n^p}{|K|}. \quad (119)$$

Thus $\text{Def}_p(K) \leq L$ is equivalent to normalized density at least e^{-L} . Two finite projection facts will be used repeatedly:

- (D1) for $1 \leq p' \leq p$, some projection of K to p' coordinates has defect at most $(p'/p) \text{Def}_p(K)$;
(D2) if P_0, P_1 partition the p coordinates and K_i is the projection to P_i , then $\text{Def}_{|P_0|}(K_0) + \text{Def}_{|P_1|}(K_1) \leq \text{Def}_p(K)$.

For (D1), multiply the sizes of all p' -coordinate projections and count how often each coordinate occurs; one projection is at least their geometric mean. For (D2), use the literal inclusion $K \subseteq K_0 \times K_1$ and take logarithms. Neither statement manufactures a new column set outside a projection of K .

For $p \geq 2$ put

$$B(p) = \left(1 + \frac{6}{5p}\right) \ln 2. \quad (120)$$

The elementary inequalities needed below are

$$B(p) \geq \ln 2, \quad B(p') \geq B(p) \quad (2 \leq p' \leq p), \quad (121)$$

$$B(\lceil p/2 \rceil) \geq \frac{\lceil p/2 \rceil}{p} (B(p) + \ln 2) \quad \text{when } p \geq 3, \quad \frac{3}{p} B(p) \leq \frac{7}{5} \ln 2 \quad (p \geq 3). \quad (122)$$

For even p , the first inequality in (122) is immediate. For $p = 2q - 1$, it reduces after clearing positive denominators to

$$\frac{6}{5}(3q - 1)(q - 1) \geq q(2q - 1),$$

which holds for $q \geq 2$. The last expression in (122) decreases with p and is largest at $p = 3$.

For $a \geq 0$ and a natural $p > 2^{a-1}$, where $2^{-1} = 1/2$, put

$$c_a(p) = p - 2^{a-1}. \quad (123)$$

The value $c_0(p) = p - 1/2$ is intentional.

Lemma 10.3 (Defect induction). *Assume*

$$3\lambda + \frac{7}{5} \ln 2 \leq -2 \ln(3/5). \quad (124)$$

Let $a \geq 0$, let $p > 2^{a-1}$, and suppose $2^a \theta \leq 1$. Every restriction of $\text{Int}_p(W)$ which retains at least A_a rows in each active copy and whose nonempty retained column set K satisfies

$$\text{Def}_p(K) \leq c_a(p)\lambda \quad (125)$$

has communication complexity at least $C_W + a$.

The induction uses the following one-bit-down statement.

Claim 10.4 (Auxiliary defect allowance). *For $a \geq 2$ and $p > 2^{a-1}$, under the same row condition, the same legality condition $2^a \theta \leq 1$, and*

$$\text{Def}_p(K) \leq c_a(p)\lambda + B(p)$$

implies communication complexity at least $C_W + a - 1$.

Proof. We prove the claim by induction on a . At $a = 2$ we have $p \geq 3$. By (D1), some projection to three copies has defect at most

$$\frac{3}{p} \text{Def}_p(K) \leq 3\lambda + \frac{3}{p} B(p) \leq 3\lambda + \frac{7}{5} \ln 2 \leq -2 \ln(3/5).$$

Its normalized column density is therefore at least $(3/5)^2$, and every one of its copies retains at least A_2 rows. The three-copy estimate (115) gives cost at least $C_W + 1$, the required value when $a = 2$. It also shows that every later restriction satisfying the auxiliary hypotheses is nonconstant.

Now take $a \geq 3$. Suppose first that Alice sends the first bit. In each active copy assign that copy to a child retaining at least $\lceil A_a/2 \rceil \geq A_{a-1}$ rows. Let p_i be the number assigned to child i , and let L_i be the defect of the projection of K to those copies. Then

$$p_0 + p_1 = p, \quad L_0 + L_1 \leq \text{Def}_p(K). \quad (126)$$

Put $d_i = \max\{p_i - 2^{a-2}, 0\}$. Since $(u - h)_+ + (v - h)_+ \geq u + v - 2h$,

$$d_0 + d_1 \geq p - 2^{a-1} = c_a(p).$$

Some child with $d_i > 0$ satisfies $L_i \leq d_i\lambda + B(p_i)$: otherwise the strict reverse inequalities, summed and combined with (126) and the monotonicity of B , would exceed the assumed defect. The induction hypothesis for the claim at level $a - 1$ gives child cost at least $C_W + a - 2$.

Suppose instead that Bob sends the first bit. One real child retains at least half of K , so its defect is at most $\text{Def}_p(K) + \ln 2$. Put $p' = \lceil p/2 \rceil$ and apply (D1) to that child. Direct substitution gives

$$\frac{p'}{p}c_a(p) \leq c_{a-1}(p'), \quad \frac{p'}{p}(B(p) + \ln 2) \leq B(p'). \quad (127)$$

The projected child therefore satisfies the auxiliary claim at level $a - 1$. In either speaker case a real child has cost at least $C_W + a - 2$; the parent has cost at least $C_W + a - 1$. This proves the claim. $\square$

Proof of Lemma 10.3. At $a = 0$, (D1) gives a one-copy projection of defect at most

$$\frac{1}{p} \text{Def}_p(K) \leq \frac{p - 1/2}{p} \lambda < \lambda.$$

It has column density greater than $e^{-\lambda} = 1 - \delta$ and at least A_0 rows, so (106) gives cost at least C_W .

At $a = 1$, the same projection shows that the current restriction is nonconstant. If Alice speaks first, assign copies to children as in the claim. For every nonempty child put $e_i = p_i - 1/2$. From $L_0 + L_1 \leq (p - 1)\lambda$, some child has $L_i \leq e_i\lambda$; the $a = 0$ case gives that child cost at least C_W . If Bob speaks first, keep a child of defect at most $\text{Def}_p(K) + \ln 2$ and project it to two copies. Its defect is at most

$$\frac{2}{p}(\text{Def}_p(K) + \ln 2) \leq 2\lambda + \ln 2 \leq -2\ln(3/5).$$

The top rung of (114) again gives child cost at least C_W . Hence the parent costs at least $C_W + 1$.

Finally let $a \geq 2$. Projection to three copies has defect at most $3\lambda \leq -2\ln(3/5)$, so (115) shows that the restriction is nonconstant. At an Alice root, the calculation (126) without the $B(p)$ allowance forces a child with $L_i \leq c_{a-1}(p_i)\lambda$; apply the induction hypothesis. At a Bob root, the denser child has defect at most

$$c_a(p)\lambda + \ln 2 \leq c_a(p)\lambda + B(p),$$

so Claim 10.4 applies at the same level a . Either way a real child costs at least $C_W + a - 1$, and the parent costs at least $C_W + a$. The induction is complete. $\square$

10.3 Entry at the requested depth

We now prove [Theorem 10.2](#). Put

$$x = 2^{-b}, \quad L = \ln(1/x) = b \ln 2. \quad (128)$$

Since $\lambda \geq \delta$, $\ln 2 < 1$, and [\(108\)](#) holds, we have

$$Q \geq \frac{80L}{\lambda}, \quad Q \geq 80, \quad Q\theta < \frac{1}{20}. \quad (129)$$

The bound $\delta \leq 1/64 < 1/60$ and the fixed logarithm estimate recorded in [Appendix B](#) imply [\(124\)](#).

Let d be the least positive integer satisfying

$$(1 - \delta)^d \leq 2^{-b}. \quad (130)$$

Such a d exists because $0 < 1 - \delta < 1$. Minimality is exactly

$$(d - 1)\lambda < L \leq d\lambda. \quad (131)$$

The lower bound $Q \geq 80$ gives $r_Q \geq 4$. For $t = 0, 1, 2$, define

$$a_t = r_Q - t, \quad p_t = 2^{a_t-1} + d. \quad (132)$$

Then $p_t > 2^{a_t-1}$.

We next prove the ceiling-safe count of good copies. For each t ,

$$(p_t - 1)m + (Q - p_t + 1)(A_{a_t} - 1) < Qm \frac{3/5}{2^t}. \quad (133)$$

Divide the left side by the positive number Qm . Since $Q = 2^{r_Q}$, use [\(131\)](#) and $(A_a - 1)/m < 2^a\theta$ to bound the quotient strictly by

$$2^{-t-1} + \frac{L}{Q\lambda} + \frac{Q\theta}{2^t}.$$

The last two terms are strictly below $(1/10)/2^t$ for $t = 0, 1, 2$: use $L/(Q\lambda) \leq 1/80$ and $Q\theta < 1/20$ separately in the three cases. Adding 2^{-t-1} gives $(3/5)/2^t$, proving [\(133\)](#). Also

$$d \leq L/\lambda + 1 \leq Q/80 + Q/80 < Q/2,$$

so every p_t is at most Q .

Let $V = \text{Int}_Q(W)$, so $R^+ = V^\top$. Fix $t \in \{0, 1, 2\}$ and an arbitrary retained member of

$$\mathcal{B}(\text{Rows}(R^+), \text{Cols}(R^+); 1, 2^{-b}, (3/5)/2^t).$$

After transposition it is a literal restriction of V with at least $\lceil Qm(3/5)/2^t \rceil$ retained rows in total and a nonempty column set of normalized density at least 2^{-b} . Call an outer copy *good* if it contains at least A_{a_t} retained rows. If fewer than p_t copies were good, its total row count would be at most the left side of [\(133\)](#), a contradiction. Hence at least p_t copies are good.

Choose p_t good copies and project the retained column set to them. The specified-coordinate projection preserves normalized density, so the new column defect is at most L . Moreover,

$$2^{a_t}\theta = Q\theta/2^t < 1, \\ L \leq d\lambda = (p_t - 2^{a_t-1})\lambda = c_{a_t}(p_t)\lambda.$$

All hypotheses of [Lemma 10.3](#) now hold, and the retained member has cost at least

$$C_W + a_t = C_W + r_Q - t. \quad (134)$$

Since the member and t were arbitrary, the three instances of (134) are exactly (112).

For $t = 0$, the full matrix is one member of the bracket family, so $D(R^+) \geq C_W + r_Q$. The explicit selector protocol sends the coordinate in r_Q bits and then runs an optimal protocol for W , giving $D(R^+) \leq C_W + r_Q$. This proves (111). Replacing the optimal-cost equality by any bound $D(W) \leq Y_W$ in the same selector protocol proves (113).

For the canonical schedule, $\delta = \delta_U$ and $\theta = 1/(160Q_{\text{br}})$. Equations (44)–(51) establish the hypotheses, and the selected width is exactly Q_{br} . Therefore the bridge output in (109) is definitionally the R^+ in (13); there is no transport between two similarly named successors.

11 One-level amplifier

We now assemble the construction-specific theorem and the generic bridge. Fix P, π, g before either source predicate is used. Let $\text{Succ}(P, \pi) = \text{Can}(R^+)$ and let H be the overhead in (34).

Theorem 11.1 (One-level amplifier). *Assume $\text{LevelOK}(\mathcal{F}, P, \pi)$ and the scalar conditions that do not mention the current source promise.*

(i) *If $\text{SourceYes}(S)$ and $D(R) \leq Y$, then*

$$D(\text{Succ}(P, \pi)) \leq Y + H. \quad (135)$$

(ii) *If $\text{NoLevelOK}(\mathcal{F}, P, \pi, g)$ holds, then*

$$D(\text{Succ}(P, \pi)) = D_0 + H + 1, \quad (136)$$

$$D_0 + H + 1 \leq \text{Rob}(\text{Succ}(P, \pi); b_+, 3/5). \quad (137)$$

(iii) *If $m_N = |\text{Rows}(N)|$ and $n_N = |\text{Cols}(N)|$, then the actual successor carriers satisfy*

$$|\text{Rows}(\text{Succ}(P, \pi))| = (q_{\text{al}} m_N)^{Q_{\text{br}}}, \quad (138)$$

$$|\text{Cols}(\text{Succ}(P, \pi))| = Q_{\text{br}} n_N^{q_{\text{al}}}. \quad (139)$$

Proof. For (i), the source colouring chooses the consistent guarded-source answers. The resulting positive-branch fourth-stage protocol does not need the final template/source distinguishing bit used by the unconditional upper protocol. Together with the explicit Stage–1, Stage–2, and Stage–3 selectors it gives

$$D(N) \leq Y + a + \log_2 q_2 + \log_2 s.$$

Equation (104) adds $\log_2 q_{\text{al}}$, and (113) adds $\text{clog}_2 Q_{\text{br}}$. The sum is $Y + H$. Reindexing preserves cost by [Lemma 3.2](#).

For (ii), [Theorem 9.3](#) gives the common-threshold input with

$$C_W = D_0 + a + \log_2 q_2 + \log_2 s + 1 + \log_2 q_{\text{al}}.$$

Apply [Theorem 10.2](#). Since its selector exponent is $\text{clog}_2 Q_{\text{br}}$, the resulting target is precisely $D_0 + H + 1$. Transfer both exact cost and profile through the canonical reindexing by [Lemma 3.3](#).

For (iii), use (15); canonical reindexing changes only the carrier names. Positivity follows from the level checklist and from positivity of powers. $\square$

What is and is not iterative

The construction in [Theorem 11.1](#) is already fixed before either implication is proved. The negative implication supplies exact cost and the same three-rung property needed as input at the next level. The positive implication supplies only an upper bound. The next section carries both statements simultaneously on one sequence; it does not construct separate positive and negative sequences.

12 The one canonical matrix-and-budget recursion

Fix a formula Φ , one source branch S , and the depth $b = b(n(\Phi))$ from (58). The source branch stays fixed throughout the recursion.

12.1 Full-cube seed

Let $a_{\text{seed}}(b)$ be the least $a \geq 3$ satisfying

$$64(b+1) \leq 2^a - 2, \quad (140)$$

and put $q_{\text{seed}} = 2^{a_{\text{seed}}(b)} - 2$. Define

$$F_b : ([q_{\text{seed}}] \rightarrow \{0, 1\}) \times [q_{\text{seed}}] \rightarrow \{0, 1\}, \quad F_b(x, i) = x(i). \quad (141)$$

Its canonical zero-based presentation has

$$|\text{Rows}(F_b)| = 2^{q_{\text{seed}}}, \quad |\text{Cols}(F_b)| = q_{\text{seed}}, \quad y_0 = a_{\text{seed}}(b) + 1. \quad (142)$$

Lemma 12.1 (Seed semantics). *For $b \geq 1$,*

$$D(F_b) = y_0, \quad y_0 \leq \text{Rob}(F_b; b, 3/5), \quad w(F_b) = q_{\text{seed}}. \quad (143)$$

Proof. Bob sends the coordinate in a_{seed} bits and Alice sends its value, so $D(F_b) \leq a_{\text{seed}} + 1$. The unit-vector rows cut out a $q_{\text{seed}} \times q_{\text{seed}}$ identity matrix inside $\mathbf{1}_{F_b=1}$. Minimality and $a_{\text{seed}} \geq 3$ give $2^{a_{\text{seed}}-1} < q_{\text{seed}} \leq 2^{a_{\text{seed}}}$. Thus [Lemma 3.1](#), with $d = a_{\text{seed}}$, gives the reverse inequality.

For the retained rungs, fix a row set A of density at least 2^{-b} . It contains at least $2^{q_{\text{seed}}-b}$ Boolean vectors, so its real span has dimension at least $q_{\text{seed}} - b$: a d -dimensional real subspace contains at most 2^d Boolean vectors. If J is the retained column set, coordinate projection has a kernel of dimension at most $q_{\text{seed}} - |J|$. Hence the restricted one-output indicator has rank at least $|J| - b$. At rung $r \in \{0, 1, 2\}$,

$$|J| \geq \lceil q_{\text{seed}}(3/5)2^{-r} \rceil, \quad |J| - b > 2^{a_{\text{seed}}-r-1},$$

where the strict inequality follows from $q_{\text{seed}} \geq 64(b+1)$. Since $a_{\text{seed}} - r \geq 1$, the restricted matrix is nonconstant and [Lemma 3.1](#), now with $d = a_{\text{seed}} - r$, gives cost at least $a_{\text{seed}} + 1 - r$. Adding the rung debt gives the profile bound. Finally, $\text{clog}_2(2^{q_{\text{seed}}}) = q_{\text{seed}}$ and $\text{clog}_2 q_{\text{seed}} \leq q_{\text{seed}}$. $\square$

12.2 Dependent state

A state at level i is a pair

$$\mathcal{X}_i = (P_i, y_i), \quad P_i = (S, m_i, n_i, R_i), \quad (144)$$

where $R_i : [m_i] \times [n_i] \rightarrow \{0, 1\}$ is the actual current matrix and y_i is its positive-branch comparison budget. Initialize with

$$R_0 = \text{Can}(F_b), \quad y_0 = a_{\text{seed}}(b) + 1. \quad (145)$$

At level i , feed the schedule the exact public values

$$I_i = (D_i, Y_i, b_0, b_+) = (y_i + i, y_i, b, b). \quad (146)$$

With the formula-dependent but promise-independent exponent $h(n)$ from (59), define

$$\pi_i = \pi(P_i, h(n), I_i), \quad g_i = g(P_i, h(n), I_i), \quad (147)$$

$$H_i = a_i + \log_2 q_{2,i} + \log_2 s_i + \log_2 q_{\text{al},i} + \text{clog}_2 Q_{\text{br},i}. \quad (148)$$

The recursion is

$$R_{i+1} = \text{Succ}(P_i, \pi_i), \quad y_{i+1} = y_i + H_i. \quad (149)$$

Equations (144)–(149) are one ordinary recursion on the dependent state. There is no separately supplied matrix chain, width list, or success trace.

Lemma 12.2 (Definitional recursion facts). *For every i :*

- (a) *the source field of P_i is S ;*
- (b) *the seed-cost field of g_i is $y_i + i$;*
- (c) *the input and output robustness depths in g_i are both b ;*
- (d) *the matrix field of P_{i+1} is the unique successor in (77);*
- (e) *the carriers of P_{i+1} satisfy (138)–(139) for the actual protected N_i .*

Proof. Each item is obtained by unfolding one recursion equation. No induction hypothesis about communication complexity is needed. $\square$

12.3 Root family

For the fixed source list $\mathcal{S}(\Phi)$, run the recursion separately for each index $j \in [t(\Phi)]$ but with the same $b(n)$, $h(n)$, and level number k . Define

$$M_{\Phi,k,j} = R_{k,j}, \quad Y_{\Phi,k,j} = y_{k,j}. \quad (150)$$

The index set is precisely the source-list index set. The source theorem proves it nonempty beyond its fixed cutoff.

13 Gap and size induction

Fix $k \geq 1$ and choose the common cutoff from Lemma 6.4. Let Φ be valid beyond that cutoff, and fix one source branch S_j . All structural and scalar conditions below are now theorems about the state recursion.

13.1 Simultaneous gap invariant

Theorem 13.1 (Canonical iteration). *For every $0 \leq i \leq k$:*

$$\text{SourceYes}(S_j) \implies D(R_{i,j}) \leq y_{i,j}, \quad (151)$$

$$\text{SourceNo}(S_j) \implies \begin{cases} D(R_{i,j}) = y_{i,j} + i, \\ y_{i,j} + i \leq \text{Rob}(R_{i,j}; b, 3/5). \end{cases} \quad (152)$$

The same sequence $(R_{i,j}, y_{i,j})$ occurs in both implications.

Proof. Induct on i .

At $i = 0$, both branches use [Lemma 12.1](#). The positive implication is its equality weakened to an upper bound; the negative implication is its exact and robust pair.

Suppose the statement holds at $i < k$. The cutoff theorem supplies [LevelOK](#) and the scalar conditions for the actual state P_i and the actual parameters π_i . On the positive source branch, combine the induction hypothesis $D(R_i) \leq y_i$ with [Theorem 11.1](#)(i):

$$D(R_{i+1}) \leq y_i + H_i = y_{i+1}.$$

On the negative source branch, the induction hypothesis and the definitional schedule equation (146) give exactly

$$D(R_i) = D_i, \quad D_i \leq \text{Rob}(R_i; b_0, 3/5).$$

Together with the unchanged source facts, these are [NoSeed](#)(P_i, g_i). Hence the full negative checklist holds. Apply [Theorem 11.1](#)(ii):

$$\begin{aligned} D(R_{i+1}) &= D_i + H_i + 1 = (y_i + i) + H_i + 1 = y_{i+1} + (i + 1), \\ y_{i+1} + (i + 1) &\leq \text{Rob}(R_{i+1}; b, 3/5). \end{aligned}$$

This proves both statements simultaneously and also proves the exact seed facts needed by the next level. No case-dependent object is introduced. $\square$

13.2 Actual carrier counts

The exact finite counts at one level are useful both for checking nonemptiness and for bounding the final matrix. Let

$$\begin{aligned} x_1 &= q_1 m_i, & y_1 &= n_i^{q_1 + s + 1}, \\ x_2 &= q_2 y_1, & y_2 &= N_2, \\ x_3 &= s y_2, & y_3 &= x_2^s. \end{aligned}$$

For the raw and reservoir counts define

$$\begin{aligned} T &= x_3, & V &= 4s v(G) m_i, & B_0 &= 4s(s - 3) m_i, \\ X &= T + V + B_0, & S &= y_3, & D' &= x_2, & M &= n_i^{s+1} S. \end{aligned}$$

Substituting the multiplicities (69)–(70) and collecting equal monomials gives the exact collapse

$$|\text{Rows}(A_i)| = 2^{22} X T V B_0, \quad (153)$$

$$|\text{Cols}(A_i)| = 184 M S D'. \quad (154)$$

Since $N_i = A_i^\top$, these two values swap. Therefore

$$m_{i+1} = (q_{\text{al},i} |\text{Cols}(A_i)|)^{Q_{\text{br},i}}, \quad (155)$$

$$n_{i+1} = Q_{\text{br},i} |\text{Rows}(A_i)|^{q_{\text{al},i}}. \quad (156)$$

These are identities for the matrix in (149).

13.3 One scalar naming-width majorant

Suppose one integer $z \geq 1$ satisfies

$$X, T, V, B_0, M, S, D' \leq 2^z, \quad q_{\text{al}}, Q_{\text{br}} \leq z. \quad (157)$$

Equations (153) and (154) give

$$|\text{Rows}(A)| \leq 2^{22+4z}, \quad |\text{Cols}(A)| \leq 2^{8+3z}. \quad (158)$$

Substitution into (155)–(156) yields

$$\text{clog}_2 m_{i+1} \leq (z + (8 + 3z))z, \quad (159)$$

$$\text{clog}_2 n_{i+1} \leq z + (22 + 4z)z. \quad (160)$$

Define

$$A(z) = \max \{1, (z + 8 + 3z)z, z + (22 + 4z)z\}. \quad (161)$$

For $z \geq 1$, the second polynomial dominates and

$$A(z) = 4z^2 + 23z \leq 27z^2. \quad (162)$$

At a level, bound the balanced-family size by (25); use it in the seven displayed carrier counts; and let z_i^{pre} be the maximum of their ceiling logarithms and the two wrapper widths. Before selecting a branch, take the maximum of z_i^{pre} over all source-list indices and all $i < k$, together with the seed naming width. Call the result z_0^* . Define the scalar sequence

$$z_0 = z_0^*, \quad z_{i+1} = A(z_i). \quad (163)$$

Theorem 13.2 (Actual naming-width induction). *For every source-list index j and every $0 \leq i \leq k$,*

$$w(R_{i,j}) \leq z_i. \quad (164)$$

If the cutoff comparison $z_k \leq e(n(\Phi), k)$ holds, then

$$w(R_{k,j}) \leq e(n(\Phi), k) \quad (165)$$

for every j .

Proof. The seed case is the definition of z_0^* . At a successor, the structural checklist and balanced-family size theorem prove (157) at the current scalar bound. The exact carrier identities then give (159)–(160), so $w(R_{i+1,j}) \leq A(z_i) = z_{i+1}$. The final statement is transitivity. The only formula-size estimate is the last numerical comparison, and the cutoff in Lemma 6.4 makes it uniform over all branches. $\square$

14 Assembly of the finite theorem

We now discharge the quantifiers in [Theorem 1.1](#). Fix the two outside contracts once and for all, and then fix an integer $k \geq 1$. Let U_k be the maximum of the source cutoff and the simultaneous cutoff in [Lemma 6.4](#). Enlarge it, if necessary, until $e(n, k) \geq 1$ whenever $n \geq U_k$. This last enlargement depends on k but not on a formula or a source-list index.

For a valid ordered formula Φ with $n(\Phi) \geq U_k$, set

$$t(\Phi) = |\mathcal{S}(\Phi)|.$$

For every $j \in [t(\Phi)]$, use the source branch $S_{\Phi,j}$ in the single recursion of [Section 12](#), and define the output record by

$$O_{\Phi,k,j} = (|\text{Rows}(R_{k,j})|, |\text{Cols}(R_{k,j})|, R_{k,j}, y_{k,j}). \quad (166)$$

The positivity proofs in the record are supplied by the cutoff theorem and the exact carrier recurrences. Thus the total-constructor fallback described in [Section 2](#) is not selected at any of these calls. Equation (166) is the definition of $(M_{\Phi,k,j}, Y_{\Phi,k,j})$; there is no later choice of a matrix.

Proof of Theorem 1.1. The source theorem gives $t(\Phi) > 0$, proving non-vacuity of the index set. The construction in (166) is already fixed for every index before either source implication is used.

Suppose first that Φ is satisfiable. Clause (S4) of [Assumption 4.1](#) supplies an index j with $\text{SourceYes}(S_{\Phi,j})$. Apply (151) at $i = k$ to the already constructed recursion:

$$D(M_{\Phi,k,j}) = D(R_{k,j}) \leq y_{k,j} = Y_{\Phi,k,j}.$$

This proves (3).

Suppose instead that Φ is unsatisfiable. Clause (S5) supplies $\text{SourceNo}(S_{\Phi,j})$ for every index j . Applying (152) at $i = k$ separately to each member of the same fixed family gives

$$D(M_{\Phi,k,j}) = D(R_{k,j}) = y_{k,j} + k = Y_{\Phi,k,j} + k.$$

This proves (4); in particular it is an equality, not only a lower bound.

For every index, the cutoff comparison and [Theorem 13.2](#) give

$$w(M_{\Phi,k,j}) = w(R_{k,j}) \leq e(n(\Phi), k),$$

independently of either source implication. Apply [Lemma 3.4](#) to this literal $R_{k,j}$ to obtain (6) and (7). This completes all conjuncts of the theorem. $\square$

14.1 Quantifier and identity audit

The proof above has the following order, written without suppressed choices:

$$(\text{source contract, balanced contract}) \longrightarrow \forall k \geq 1 \exists U_k \forall \Phi \exists t > 0 \exists O \quad [\text{three conclusions about } O].$$

The output function $O : j \mapsto O_{\Phi,k,j}$ occurs outside both source implications. The satisfiable case selects only an index; the unsatisfiable case selects no data. The naming-width, side, and entry statements all mention the matrix field of that same output record. These three observations rule out the most common quantifier, promise-dependence, and same-object errors.

15 Mathematical status and formal verification boundary

This manuscript is intended to be read in two different ways. As a mathematical document, it gives the complete finite construction and proof relative to the two assumptions in [Section 4](#). As a guide to a proof assistant, it is a specification whose individual declarations must still be checked against the kernel. Those two claims must not be conflated.

15.1 What is proved on paper

The human-level proof contains the following closed chain.

- (H1) The source list and balanced families are chosen without a source promise.
- (H2) Every carrier, finite map, entry rule, wrapper, and fallback is explicitly defined.
- (H3) The protected matrix has the dense exact bound and the three actual reference rungs.
- (H4) The concrete scale-alignment game excludes every shallow protocol on each retained member.
- (H5) The common-threshold bridge adds exactly one bit and returns the three-rung seed property.
- (H6) One dependent recursion proves the additive gap by induction.
- (H7) Exact carrier identities yield a scalar width recurrence and the same-matrix side and entry bounds.

The only assumed mathematics is the composed source theorem and the balanced-family contract, including its three named product clauses. The source theorem may have been proved using a finite expander result; no expander object or parameter occurs after the source list has been supplied.

15.2 What a completed Lean verification must show

A declaration corresponding to [Theorem 1.1](#) counts as formally verified only after all of the following have happened in one repository state:

- (V1) the finite root imports and compiles successfully;
- (V2) every theorem used by the root has no placeholder proof;
- (V3) the printed axiom set contains only ordinary logical principles and the two approved external theorem interfaces;
- (V4) the checked statement has the same quantifier order, additive equality, and same-matrix size clauses as [Theorem 1.1](#);
- (V5) the repository’s verification receipt records the command, source revision, and successful output.

At the date printed on this manuscript, the formal development contains many checked components of this chain. This manuscript deliberately makes no claim that the final root has passed (V1)–(V5). The current verification receipt, rather than this sentence, is authoritative and must be updated when the root is checked.

No hidden proof object

Neither a successful build of this PDF nor the presence of a Lean declaration is evidence for the final theorem by itself. The evidence is a successful kernel check of the finite root together with its axiom report. Conversely, the two external assumptions are stated openly and are not defects in that report.

A The concrete stopping argument

This appendix expands the two protocol-frontier arguments used in [Sections 8](#) and [9](#). Its purpose is to identify the exact finite objects that a formal proof must carry. It does not posit an arbitrary trace, a callback, or a terminal conclusion.

A.1 Reached-node data

Fix an actual retained row set R_C , an actual retained column set C_C , and an actual protocol Q for the corresponding restriction. A reached node is represented by the tuple

$$\mathcal{V} = (v, d, \rho, R_v, C_v, Q_v, J_v, E_v, \omega_v, \lambda_v), \quad (167)$$

with the following meanings.

- (R1) v is a node of the given protocol. The word $\rho \in \{0, 1\}^d$ begins with its unique actual root-to- v route. Only for an early Alice-only leaf in the alignment proof may ρ end with a fixed virtual suffix; then v remains the original leaf.
- (R2) $R_v \subseteq R_C$ and $C_v \subseteq C_C$ are exactly the inputs consistent with the actual part of ρ ; Q_v is the literal subtree rooted at v .
- (R3) J_v is the finite set of routed certificate indices placed at v . Each index stores its physical alignment coordinate and its clean-or-split tag; $E_v \subseteq J_v$ is the already charged exceptional subset.
- (R4) $\omega_v(j)$ is the surviving ordinary reference-row fibre of the coordinate stored by certificate j ; $\lambda_v(j)$ is the surviving foreign or source-owned fibre used by the applicable extension call.

The tuple stores subsets and maps only. The facts that these subsets have enough mass and that Q_v must be hard are proved propositions about the tuple, not fields used to construct it.

At the root, ρ is empty, $R_v = R_C$, $C_v = C_C$, $Q_v = Q$, and J_v contains the one starting reference certificate for every physical coordinate outside the projection-cover exceptional set. This proves initialization by unfolding definitions.

At an Alice node with query function $a : R_v \rightarrow \{0, 1\}$, define

$$R_{v\varepsilon} = \{r \in R_v : a(r) = \varepsilon\}, \quad C_{v\varepsilon} = C_v \quad (\varepsilon = 0, 1).$$

Every row and every fibre is partitioned between the two children. The selected child is the one supplied by the cleared mass inequality; it is an actual child of Q . At a Bob node with query $b : C_v \rightarrow \{0, 1\}$, define the two column children analogously, leaving the row set unchanged. Projection balancedness identifies an actual child in which the required column pattern survives. These are the only transition rules.

The protected proof stops at its two concrete construction events described below. The alignment proof stops at the first Bob node before its fixed Alice-prefix depth, or at a route token of exactly that depth. An early leaf receives a fixed virtual suffix solely to make the latter token; the protocol and the leaf are unchanged. Hence no stopping token properly prefixes another. The reached row sets partition R_C , and

$$\sum_{v \in \mathcal{T}} 2^{-|\rho_v|} \leq 1. \quad (168)$$

Both statements follow by induction on the finite protocol tree.

A.2 The four fixed extension corollaries

The inherited *Localised extension* corollary of [2], together with its two-copy and odd-copy lower bounds, is used through four substitution rules. The parameters (v, K, h, d) record the number of coupled streams, family width, retained mass, and already paid debt.

Call	Use	v	(K, h)	debt
(i)	protected Alice stop	1	(κ, x_{Dom})	1
(ii)	ordinary/foreign terminal pair	2	$(\kappa, 49/64)$	1
(iii)	early protected donor mass	2^{e-1}	$(s\kappa q_2, \rho_1 2^{-(2^e)-1})$	0
(iv)	exceptional final half-block	$2^{e_{\text{exc}}-1}$	$(2sq_2, \rho_0 2^{-(2^{e_{\text{exc}}})-1})$	0

At the selected protocol node v of path length d_v , set $\kappa = 2^{r_2-d_v}$ in (i)–(ii), and $\kappa = 2^{r_{\text{al}}-d_v}$ in (iii). These are node-local residual capacities, not fixed Stage–2 widths. In (i)–(ii), use localised extension for $\kappa \geq t_2$ and the inherited two-copy or odd-copy bounds for $2 \leq \kappa < t_2$. At $\kappa = 1$, positive witness excess gives either an unsplit hard witness or two ordinary coordinates; the first uses its witness lower bound and the second the two-copy bound. With N_2 the indexed Stage–2 family size, define

$$T_0^{\text{aux}} = \lceil 999N_2/1000 \rceil, \quad T_3^{\text{aux}} = \lceil 2^{1-b_T} N_2 \rceil, \quad x_{\text{Dom}} = \frac{T_0^{\text{aux}} - (s-1)T_3^{\text{aux}}}{N_2}. \quad (169)$$

This is a lower bound on the dominant-block fraction after reserving T_3^{aux} rows in each other block. Here $e \geq 1$ is the exponent of the dyadic class selected at a heavy first-Bob node, and $e_{\text{exc}} \geq 2$ is the canonical exceptional exponent from clause (B3). Thus the two ambient donor widths are $u = 2^e$ and $u_{\text{Don}} = 2^{e_{\text{exc}}}$, while the selected half-blocks have literal cardinalities 2^{e-1} and $2^{e_{\text{exc}}-1}$. Set $\rho_0 = 99/100$ and $\rho_1 = \rho_0/2 = 99/200$: the selected donor projection retains at least fraction ρ_0 , and one child after Bob’s query retains at least ρ_1 . These two fractions are fixed before the protocol; the capacity κ and selected dyadic class may depend on it. The table fixes substitution rules; the main proof never receives a function that manufactures extension conclusions.

For each row of the table the proof obligation has the same four parts: the relevant carrier maps are in range; the selected product family has the required balanced count; the displayed retained mass meets the localized extension threshold; and the incoming seed rung supplies the stated debt. The structural checklist proves the first part, the appropriate one of (B1)–(B3) proves the second, Appendix B proves the third, and (19) proves the fourth. The inherited localized extension theorem then yields a restriction of the literal current matrix.

A.3 Protected stopping

In the protected game, initialize (167) with the actual retained member of N . Alice transitions conserve the labelled template, vertex, and blocker masses. Bob transitions keep the real child selected by the ordinary balanced count. If the route exposes every selector bit, the physical selector and diagonal maps embed the required incoming seed rung. If Bob speaks first, call (i) embeds the full multistage restriction in that real child. Source synchronization proves that at least one of these events has enough uncharged mass: otherwise the routed answers define a colouring with fewer than $\gamma e(G)$ monochromatic edge occurrences. In either case the suffix costs at least $C - |\rho_v|$, contradicting the suffix of a protocol of cost below C .

A.4 Scale-alignment stopping

For the alignment game, the row partition and (84) first remove coordinates with excessive physical loss; the same aggregate estimate controls reference-fibre loss, and the product-cover theorem removes E_0 . A first-Bob node with small residual dyadic capacity is contradicted by one of the three actual reference rungs. A heavy first-Bob node has a positive dyadic exponent e and is contradicted by call (iii), using product clause (B1). In both cases the selected real Bob child costs at least $C_W - |\rho_v| - 1$, so the Bob node itself costs at least $C_W - |\rho_v|$.

Every remaining clean final certificate has exactly one owner token, and distinct clean certificates have distinct owners. Physical protected rows from its coordinate need not follow that certificate: their mass is counted at every first-Bob and final token. The four-category loss table proves that each good donor sends more than $\eta_R/2$ of normalized protected mass away from its owner.

For an owner i , let ε_i be the literal deficit of its ordinary baseline at its owner and put

$$\delta_i = q_2 2^{1-b_1} + 5\varepsilon_i + 2/q_2.$$

The disjoint owner baselines give $\sum_i \varepsilon_i \leq \eta_R/(8g)$, where $g = 91/92$. For each ordered pair of distinct good donors (j, i) , clause (B2) and call (ii) prove the following implication for a fresh physical restriction: if donor j sends mass at least $2\mu\delta_i$ to the owner of i , then that owner suffix has cost at least C . Its actual suffix budget is only $C - 1$, so every such transfer is strictly smaller than $2\mu\delta_i$. Summing these pointwise upper bounds and using the three owner gates gives total mass at other good owners at most $3\eta_R/16$. Hence every good donor sends more than $\eta_R/4$ to exceptional final tokens.

A double pigeonhole selects one exceptional token T_* , one of the two protected reservoir types, and many good donors which have that reservoir at density at least the uniform δ_T . The exact terminal gates $4E_T u_T B_{\text{pair}} < q_{\text{al}}$ and $u_T \delta_T \geq 8s$ leave u_T donors after the product-cover loss. Choose a literal half-block of size $u_T/2$. Clause (B3), whose exponent is the fixed number $e_{\text{exc}} = \log_2 u_T$, and call (iv) construct the actual exceptional terminal restriction at T_* . It has suffix cost at least C , contradicting the actual final-token budget $C - 1$.

The consumer map is therefore exact: (i) is used only by the protected dense proof; (iii) only by heavy first-Bob nodes; (ii) only by pairwise terminal transfer; and (iv) only by the exceptional terminal. No other extension interface is required.

B Arithmetic gates in consumption order

This appendix records the numerical facts which are easy to reverse or apply to the wrong level. All variables below are the canonical values for one already fixed state.

B.1 Positivity and Stage-2 range

From $h \geq 5$ we have $\sigma \geq 7$, $s = 2^\sigma \geq 128$, and therefore $s - 3 > 0$. The common source facts give $v(G), e(G), \Delta > 0$ and $\gamma > 0$, so $\zeta, \delta_V, \eta_R, \eta_C$, and δ_U are positive. The maxima in the schedule make every exponent nonnegative and every displayed power of two positive.

The active guarded-source coordinate count is at most $12se(G)$. Hence

$$\text{rank}(\alpha) < 12se(G) \leq 2^{\text{clog}_2(12se(G))} = q_2.$$

The selected stage condition $k_2 + 2 \leq r_2$ gives $4t_2 = 2^{k_2+2} \leq 2^{r_2} = q_2$. Thus $1 \leq t_2 \leq q_2$, and the ordinary balanced-family request is legal.

B.2 Ownership of the five quantitative rows

The structural readiness theorem does not store the rows in (57). Their exact proof ownership is:

- (Q1) the hard-seed cutoff proves $m_{\text{HS}} \leq h$, and the formula $\sigma = \max\{6, h + 2\}$ then proves $m_{\text{HS}} + 2 \leq \log_2 s$;
- (Q2) the source-width comparison proves $64sq_1/\zeta \leq q_2$;
- (Q3) the terminal-width estimate proves $u_T \leq q_{\text{al}}$;
- (Q4) the input-depth estimate proves $a + 2 + \sigma \leq b_0$;
- (Q5) the literal reservoir-count estimate proves $100(V_0 + B_0) \leq T_0$.

None of these five statements is treated as a definitional equality. After (Q5), the actual template density is at least $91/92$. Together with the formula-owned low-density estimate, this implies all three literal ceiling gates

$$\frac{9(s/2^j)}{16} \leq \left\lceil \frac{s(\beta_j - d_j)}{1 - d_j} \right\rceil \quad (j = 0, 1, 2), \quad (170)$$

where β_j, d_j are defined below. Thus the reference gates are derived consequences of the five rows and the schedule formulas, not three additional fields silently bundled into arithmetic readiness.

B.3 Protected loss inequalities

The factors $1/4$ in (48) and (49) give, after multiplying by the positive denominators,

$$\eta_R < \frac{(1/4)(1/184)\zeta}{8s}, \quad (171)$$

$$\eta_C \leq \frac{1}{2} \min \left\{ \frac{2^{-21}}{1000s}, \frac{2^{-22}\delta_V}{16}, \frac{2^{-22}}{16(s-3)} \right\}. \quad (172)$$

The two remaining defect conditions reduce to $8(2^a - 2) \leq 2^{k_2}$ and $64s(2^a - 2) \leq \zeta 2^{r_2}$. They are among the finite comparisons dominated by k_2 and r_2 in the canonical schedule. No matrix lower bound is used to establish them.

B.4 Reference-block ceiling

For $j \in \{0, 1, 2\}$ define

$$w_j = s/2^j, \quad T_j = 9w_j/16, \quad (173)$$

$$d_j = w_j 2^{-b_T}, \quad \beta_j = \frac{(3/5)/2^j + \tau_{\text{act}} - 1}{\tau_{\text{act}}}, \quad (174)$$

where τ_{act} is the exact template-row density of the protected duplication:

$$\tau_{\text{act}} = (1 - 2^{-20}) \frac{T}{T + V + B_0} + 2^{-22}.$$

The canonical reference-bin gate is the literal inequality

$$T_j \leq \left\lceil \frac{s(\beta_j - d_j)}{1 - d_j} \right\rceil \quad (j = 0, 1, 2). \quad (175)$$

In the names used in the construction,

$$\begin{aligned} \beta_j &= \text{relativeTemplateDensity}(3/5, \tau_{\text{act}}, j), \\ d_j &= \text{referenceRowDensity}(s, j, b_T), \end{aligned}$$

and (175) is exactly the gate named `referenceBlockTarget`(s, j). The gate is used only after $d_j < 1$, which follows from $b_T > \log_2 s$. The extremal-bin calculation then proves that a retained set of density $(3/5)/2^j$ contains at least T_j actual template blocks after deleting an $s2^{-j-b_T}$ row fraction. The ceiling is on the entire quotient; moving it to the numerator would be a different and insufficient statement.

B.5 Alignment and bridge

The definition $\delta_U = \eta_R/(16q_{\text{al}})$ gives $\delta_U q_{\text{al}} = \eta_R/16$ exactly. The selected parameter bound also gives $\delta_U \leq 1/64$. From $Q_{\text{br}} = \text{Qwidth}(b_+, \delta_U)$ and $\theta = 1/(160Q_{\text{br}})$ we obtain

$$80b_+ \leq Q_{\text{br}}\delta_U, \quad Q_{\text{br}}\theta = 1/160 < 1/20. \quad (176)$$

All denominators are positive. The rational margin used by the generic bridge is stated using $\lambda_U = -\ln(1 - \delta_U)$. Since $\delta_U \leq 1/64 < 1/60$, monotonicity of $-\ln(1 - x)$ and the fixed numerical logarithm estimate give

$$3\lambda_U + \frac{7}{5} \ln 2 \leq -2 \ln(3/5). \quad (177)$$

This is the exact analytic inequality consumed by (124); it is not replaced by a base-two logarithm or by an unchecked decimal evaluation.

B.6 Owner transfer and exceptional terminal

Put $g = 91/92$ and

$$\lambda_0 = q_2 2^{1-b_1}.$$

The three scalar rows consumed by (98) are

$$2\mu q_{\text{al}} \lambda_0 \leq \frac{\eta_R}{16}, \quad \mu \leq \frac{g}{20}, \quad \frac{64\mu q_{\text{al}}}{\eta_R} \leq q_2. \quad (178)$$

For transcription, the nonconstant two rows follow from the cleared natural comparisons

$$\eta_R^{-1} \leq q_{\text{al}}, \quad 4q_{\text{al}}^2 \leq q_2, \quad q_2^2 \leq 2^{b_1}.$$

Indeed the first and third comparisons give $64\mu q_{\text{al}}/\eta_R \leq q_2$ because $64\mu \leq 1$; multiplying once more by $q_2/2^{b_1}$ gives the first row of (178). The middle row is the fixed rational calculation $1/184 \leq (91/92)/20$. These comparisons are formula-owned consequences of the alignment and Stage-2 exponents, not assumptions about a protocol.

Let B_{pair} , E_T , δ_T , u_T , and t_{ref} be the literal values in (53). The terminal projection calculation gives

$$8 \max\{1, E_T\} \leq q_{\text{al}}, \quad 4E_T u_T B_{\text{pair}} < q_{\text{al}}, \quad 4t_{\text{ref}} \leq q_{\text{al}}. \quad (179)$$

To see the strict middle comparison without real-number rounding, put $K = k_{\text{ref}} + \sigma$. The schedule bounds

$$E_T \leq 2^{6K+51}, \quad u_T \leq 2^{8K+58}, \quad B_{\text{pair}} \leq 2^{K+15}.$$

Consequently $4E_T u_T B_{\text{pair}} \leq 2^{15K+126}$, while the definition $r_{\text{al}} = 64(k_{\text{ref}} + L_{\text{ref}} + \sigma + 1)$ gives $15K + 126 < r_{\text{al}}$ and hence the strict middle row. The same exponent comparison gives the other two rows.

Finally, the power-of-two rounding in the definition of u_T gives

$$u_T \delta_T \geq 8s. \quad (180)$$

All quantities in (179)–(180) are fixed before the alleged protocol is inspected. In particular $e_{\text{exc}} = \text{clog}_2 u_T$ is the fixed exponent used by product clause (B3), and $2^{e_{\text{exc}}} = u_T$.

B.7 Carrier and final-size arithmetic

The exact protected multiplicities collect to

$$|\text{Rows}(A)| = 2^{22} XTV B_0, \quad |\text{Cols}(A)| = 184 MSD'.$$

Under the common envelope (157), use $2^{22} \leq 2^{22}$ and $184 < 2^8$ to obtain $|\text{Rows}(A)| \leq 2^{22+4z}$ and $|\text{Cols}(A)| \leq 2^{8+3z}$. The two actual successor equations then yield the exponents in (159) and (160); for $z \geq 1$ their maximum is $4z^2 + 23z \leq 27z^2$.

Finally, with $e = \lfloor n/(2k) \rfloor$,

$$e(2k) \leq n. \quad (181)$$

For $k > 0$, (181) implies $2e \leq \lfloor n/k \rfloor$. This is the only arithmetic needed to pass from the square side cap to the public entry cap.

C Dependency audit

The following table records the mathematical proof graph. “Outside” means an assumption of Theorem 1.1; “proved here” means a theorem of this manuscript, even if its Lean counterpart is still being checked.

Node	Input	Output	Status
F0	finite matrices and protocols	transports, restrictions, interlaces, three-rung profile	proved here
F1	public state and canonical schedule	structural and scalar checklists	proved here
F2	F0, F1, source branch, balanced-family data	literal Stage-1-4 matrices $B, A, N, N_{\text{ref}}, U, W, R^+$	proved here
F3	F2, negative seed facts, three product clauses	protected and scale-aligned exact lower bounds	proved here
F4	F0 and the common-threshold interface	one-bit bridge with restored three-rung profile	proved here
F5	F1–F4 and the full-cube seed	simultaneous k -level positive/negative induction	proved here
F6	exact F2 carrier counts and balanced-family size	actual-chain naming-width bound	proved here
F7	source semantics, F5, F6	finite root theorem and same-matrix entry bound	proved here
S	ordered formula	finite nonempty source list with common, positive, and negative clauses	Outside
B	(q, t, m) and the three selected product shapes	finite family, ordinary balance, and product balance	Outside

The expander used in one proof of node S is not a third arrow into F1–F7. Once S has returned the source list and its five properties, every downstream definition is independent of that proof and of any expander parameter.

C.1 Public-interface correspondence

The stable public objects have direct proof-assistant counterparts. The names below are locating evidence only; the mathematical meanings were defined before these names are mentioned.

Mathematical object	Lean-facing interface
ordered finite source list	<code>FiniteSourceSpec</code>
finite balanced-family provider	<code>FiniteBalancedFamilySpec</code>
three selected product clauses	<code>CanonicalExternalBalancednessAt</code>
public matrix state	<code>PublicInput</code>
six-number level data	<code>LevelParameters</code>
structural checklist	<code>LevelChecklist</code>
negative and scalar checklist	<code>LevelSemanticChecklist</code>
literal successor	<code>FiniteTotalLevelConstruction.successor</code>
one actual dependent chain	<code>actualPublicInputChain</code>
branch output and size clauses	<code>BranchOutput, FiniteRootSpecFor</code>

The internal lemma decomposition is allowed to differ between paper and Lean. The following may not differ: the two outside assumptions, the actual matrix constructor, the source-list quantifiers, the k -bit equality on the negative side, and the side and entry bounds on that same matrix.

D Exact outside and inherited contracts

This appendix restates the interfaces in a form suitable for direct theorem parameters. It separates genuine outside mathematics from ordinary communication lemmas proved once and reused.

D.1 Composed source interface

The source parameter consists of constants $(c_{\text{src}}, \Delta, n_{\text{src}}, \gamma)$ and one function $\Phi \mapsto \mathcal{S}(\Phi)$. Its proposition has exactly the following fields, for every valid Φ beyond n_{src} :

- source.1** the ordered list $\mathcal{S}(\Phi)$ is nonempty;
- source.2** every indexed graph has positive loop-free bounded-degree common data;
- source.3** every indexed graph has at least $n(\Phi)$ edge occurrences and obeys the displayed source-size envelope;
- source.4** satisfiability gives an index whose graph is properly three-colourable;
- source.5** unsatisfiability makes every indexed graph γ -far from every three-colouring.

The output list in all five fields is definitionally the same list. The interface contains no communication matrix and no conclusion of the present theorem. A proof of this composed source interface may invoke a finite expander theorem internally. The downstream parameter list does not contain that expander, its vertices, or its expansion constant.

The input may initially use the real logarithm in (S3). [Lemma 4.2](#) converts it to the implementation's integer-logarithm envelope. The conversion doubles the fixed exponent, raises the cutoff to at least 2, and leaves every graph and the actual matrix construction unchanged; no extra source hypothesis is added.

D.2 Balanced-family interface

The base provider maps each triple (q, t, m) to an explicitly indexed family $\mathcal{F}(q, t, m)$. Under $m > 0$ and $1 \leq t \leq q$, the base proposition contains exactly nonemptiness, entry range, the cardinality bound (25), and the denominator-cleared pattern count (26).

At each canonical level, the additional outside proposition has exactly three fields:

firstBobProducts

For every $e \geq 1$ and every $J' \subseteq [2^e]$ with $|J'| = 2^e/2$,

$$\text{Bal}_{t_2}(\mathcal{C}^{\times D_{e,J'}}; \epsilon_2).$$

The quantifier over e is outside the implication describing a heavy node; the field therefore can be specialized after the frontier chooses its dyadic class.

terminalTwoFold

$$\text{Bal}_{t_2}(\mathcal{C}^{\times [2]}; \epsilon_{\text{prod}}).$$

exceptionalProducts

For the canonical $e_{\text{exc}} \geq 2$ and every $J' \subseteq [2^{e_{\text{exc}}}]$ with $|J'| = 2^{e_{\text{exc}}}/2$,

$$\text{Bal}_{t_2}(\mathcal{C}^{\times D_{e_{\text{exc}}, J'}}; \epsilon_2).$$

Here $\mathcal{C}$, $D_{e,J'}$, ϵ_2 , and ϵ_{prod} are the literal objects in (24) and (27); there is no existentially selected replacement family. The finite equivalence used by the localized extension call sends its ambient coordinate carrier to $D_{e,J'} \times [q_2]$. Under this equivalence its entry at (d, α) is definitionally $c_{\lambda(d), \alpha}$, so the displayed family is the one read by the constructed matrix.

These are finite counting assertions. They do not quantify over a protocol, do not mention communication cost, and do not assert that a terminal node is hard. Those conclusions are derived by the four concrete extension calls in Appendix A.

[Theorem 4.4](#) constructs one provider satisfying the base and all three product clauses at the original size bound. Its finite-field and bucket-counting proof is external to Lean, not an unproved strengthening of base balancedness. The theorem parameters expose the result of that proof; they do not assert that every base-balanced provider has the same product guarantees.

D.3 Inherited communication lemmas

The following are reusable theorems, not additional outside assumptions of the root.

- (I1) Bijections of row and column carriers preserve entries, deterministic communication cost, retained-member minima, and the three-rung profile.
- (I2) Labelled duplication preserves deterministic communication cost; transposition preserves cost and swaps row and column retention.
- (I3) The selector protocol gives $D(\text{Int}_q(M)) \leq D(M) + \log_2 q$ when q is a positive power of two.
- (I4) The full-cube seed has the exact cost and three retained-rung property in [Lemma 12.1](#).
- (I5) The localized extension theorem, specialized only through calls (i)–(iv), converts actual balanced product counts and an incoming seed rung into a physical hard restriction.
- (I6) The common-threshold bridge of [Theorem 10.2](#) converts its three displayed hypotheses into exact cost and the restored three-rung property.

Every use in this manuscript names the actual carrier maps and checks the positivity hypotheses before applying these theorems.

E Adversarial proof audit

This final audit is part of the proof specification. A negative answer to any item invalidates the claimed finite theorem.

- (A1) **Non-vacuity.** The source list is nonempty, every branch output has positive row and column carriers, and every retained family minimized over in a profile is nonempty.
- (A2) **Quantifier order.** The outside contracts precede k ; the cutoff precedes Φ ; the entire output family precedes both source implications.
- (A3) **No promise-dependent constructor.** Neither a colouring nor a far-from-colourable proof is an argument of a matrix constructor or the parameter schedule.
- (A4) **No conclusion-bearing data.** The public input, level data, gap parameters, and branch output contain no communication lower bound, retained-family bound, successful trace, or final size assertion.

- (A5) **Literal Stage–4 identity.** The raw, protected, reference, aligned, transposed, and bridge matrices in the lower-bound proof are the objects defined in [Section 7](#); no look-alike matrix is substituted.
- (A6) **Reference identity.** N_{ref} uses the actual reference rows and all base copies of all raw columns, so its column carrier is exactly the column carrier of N .
- (A7) **Density order.** Every transpose swaps the two retention densities, and every profile rung uses $(3/5)/2^r$ with debt r for $r = 0, 1, 2$.
- (A8) **Exact negative cost.** Each lower bound is paired with the explicit selector upper protocol before equality is asserted. Iteration therefore adds exactly one bit per level.
- (A9) **Actual recursion.** The matrix chain is a single dependent recursion. There is no free matrix sequence, free width list, or supplied success certificate.
- (A10) **Natural subtraction and powers.** Positivity and lower bounds are established before truncated subtraction, finite indexing, logarithm, or division is used. Exact powers of two are proved by identities rather than inferred from positivity.
- (A11) **External boundary.** The only root assumptions are the composed finite source interface and the balanced-family interface with its three product fields. An expander may occur inside a proof of the former, but it is not a downstream parameter.
- (A12) **Same matrix.** The width induction uses the exact carriers of $R_{k,j}$, and the side and entry inequalities are consequences about that same $R_{k,j}$.
- (A13) **Axiom report.** Formal completion requires the checked finite root to print no hidden project assumption or opaque theorem carrying the desired conclusion.

Most dangerous confusions

The dense lower bound on N is not a low-density full-cost bound on N . The low rungs belong to the actual reference restriction and are aligned only after interlacing and transposition. Balanced product counts are not protocol lower bounds. A compiled PDF is not a checked formal theorem. Finally, a size bound for a numerical surrogate is not enough: the carrier equalities must be proved for the literal output matrix.

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